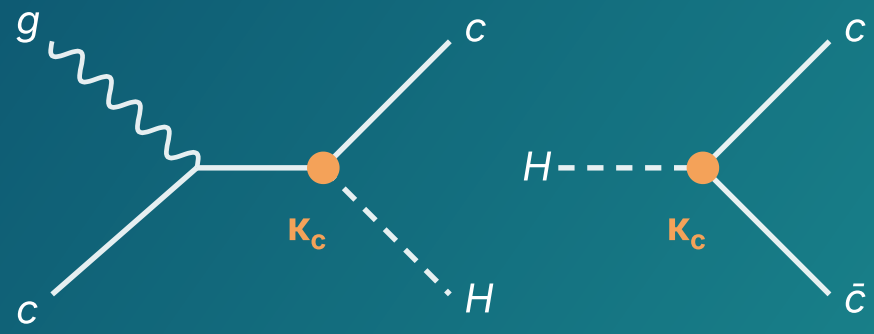
# Constraints on Higgs–charm couplings at CMS

*Higgs 2026 · Louvain-la-Neuve · 28 October 2026*

**Chirayu Gupta**
*Vrije Universiteit Brussel*

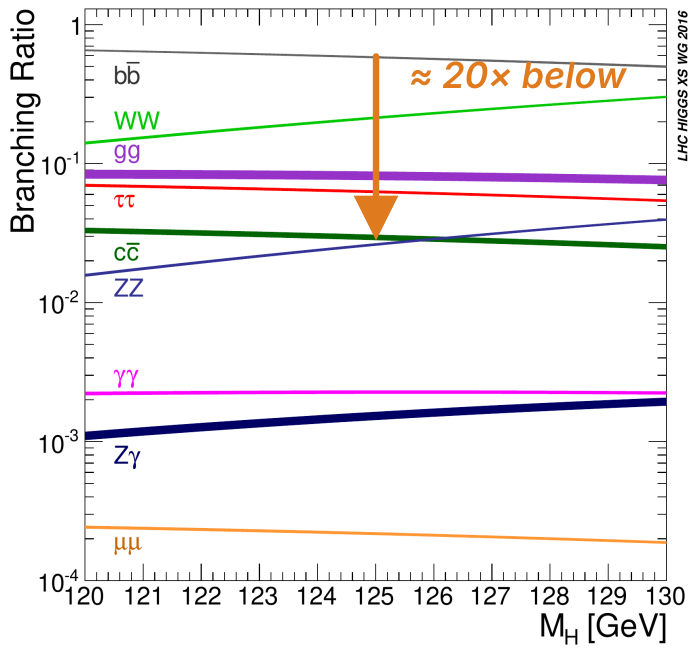
on behalf of the CMS Collaboration



# Why the Higgs–charm coupling?

## Second-generation quark Yukawas are the open frontier

- Since 2012: couplings to **t, b, τ** measured, all SM-like
- Evidence for **H → μμ** (backup)
- **Charm is the most accessible second-generation quark**



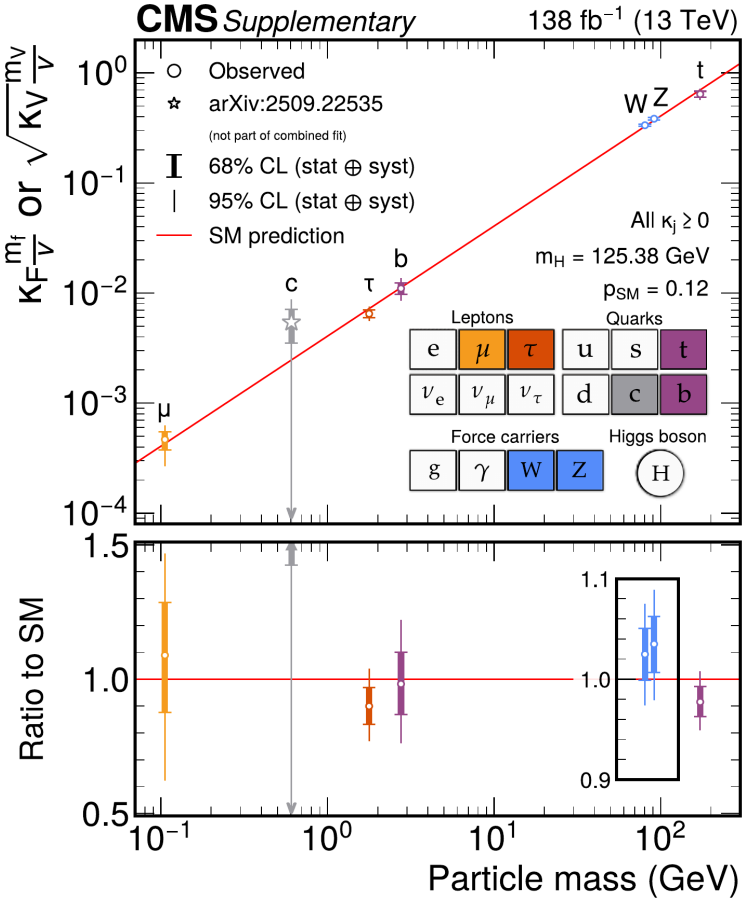

$\mathcal{B}(H \to c\bar{c}) \approx$ **2.9%**, about **20× below** $b\bar{b}$, with huge QCD backgrounds and hard c tagging (slide 3)

$$\kappa_c = y_c / y_c^{SM} \quad \text{(SM: 1)}$$

Flavour-compatible BSM, e.g. spontaneous flavour violation, allows $|\kappa_c| \sim 3$: **current sensitivity is entering this region**

SFV: PRL 123 (2019) 031802
$\kappa_c$ in SFV: JHEP 02 (2025) 067

LHC Higgs XS WG, Yellow Report 4, arXiv:1610.07922




CMS, Rep. Prog. Phys. 89 (2026) 087801 (suppl.); $\kappa_c$ point: separate $t\bar{t}H$+VH result, not in the fit

***➜ Does the Higgs give charm its mass?***

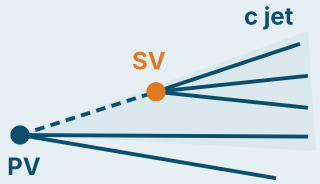
CMS-DP-2025-081

# Charm tagging

## Finding c jets between b and light jets

### Charm signature

c-hadron mass and lifetime lie **between light and b**: a secondary vertex (SV) like b jets, but closer to the primary vertex, so **easily mistaken for light jets**




### Algorithm evolution at CMS

**Likelihood / BDT**

**JP**
likelihood of track impact parameters

**CSV, CSVv2**
likelihood ratio, then MLP

**Run 2 c tagger**
BDT: CvsL, CvsB

**Deep NN**

**DeepCSV**
dense DNN (4 × 100), b/c/light classes

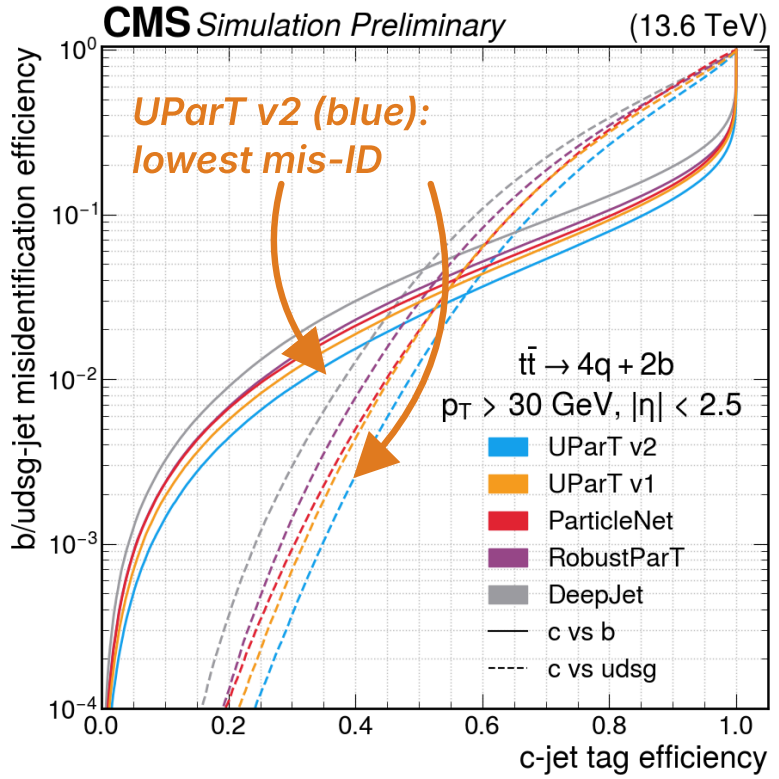
**DeepJet**
1×1 convolutions + LSTMs on all jet constituents

**Particle cloud**

**ParticleNet**
graph NN (dynamic graph CNN)

**UParT**
transformer with pairwise interaction features, adversarial training

*Run 1 → Run 3: likelihoods and trees → dense nets → graphs and attention*




$t\bar{t}$ simulation, 13.6 TeV · solid: c vs b, dashed: c vs light

In this talk **DeepJet · VH resolved, cH** **ParticleNet · VH merged, $t\bar{t}$H** **DeepDoubleX · boosted H**

# How CMS probes $\kappa_c$

## In the decay, in production, and through the Higgs width

### Direct $H \to c\bar{c}$

slides 5–11

rate $\propto \kappa_c^2/\Gamma_H$ · VH, $t\bar{t}H$, boosted (mostly ggH)

**$|\kappa_c| < 3.5$ (2.7)** $t\bar{t}H + VH$

PRL 131 (2023) 061801 · PRL 136 (2026) 011801 · PRL 131 (2023) 041801

### H + charm production

slides 12–14

cg → cH: $\kappa_c$ in the production vertex · H → γγ, WW

**$|\kappa_c| < 47$ (51)**

JHEP 11 (2025) 060 · JHEP 08 (2026) 051

### Indirect: Higgs rates and width

slide 15

on-shell and **new** off-shell H → ZZ → 4ℓ

**$\kappa_c \in [-3.3, 2.8]$ … $[-8.6, 7.5]$** by assumptions

PRD 112 (2025) 112001 · arXiv:2607.23352

### Rare exclusive decays

slide 15

H → J/ψ γ, ψ(2S) γ: sensitive to the *sign* of $\kappa_c$

**$-166 < \kappa_c < 208$**

PLB 865 (2025) 139462

$\mu = \sigma\mathscr{B}/(\sigma\mathscr{B})_{SM}$ · $\Gamma_H$ = total Higgs width · μ limits and $\kappa_c$ intervals at 95% CL, observed (expected)

***→ All $\kappa_c$ results: Run 2 (13 TeV, up to 138 $fb^{-1}$) · Run 3 so far: better triggers (slide 17)***

# VH(H → cc̄): strategy

PRL 131 (2023) 061801 · CMS-HIG-21-008

## Cleanest signature: the W or Z decays to leptons

**0ℓ · Z → νν** **1ℓ · W → ℓν** **2ℓ · Z → ℓℓ**

Main backgrounds V+jets and tt̄, normalised in control regions

*below* $p_T^H = 300$ GeV *above*

### Resolved · two small-R jets (R = 0.4)

- **>95% of the signal**, but more background
- **DeepJet** CvsL and CvsB: the two highest-CvsL jets form the H candidate
- The Higgs mass peak is up to **20% sharper**, thanks to a c-jet energy regression, recovery of FSR jets and, in 2ℓ, a kinematic fit

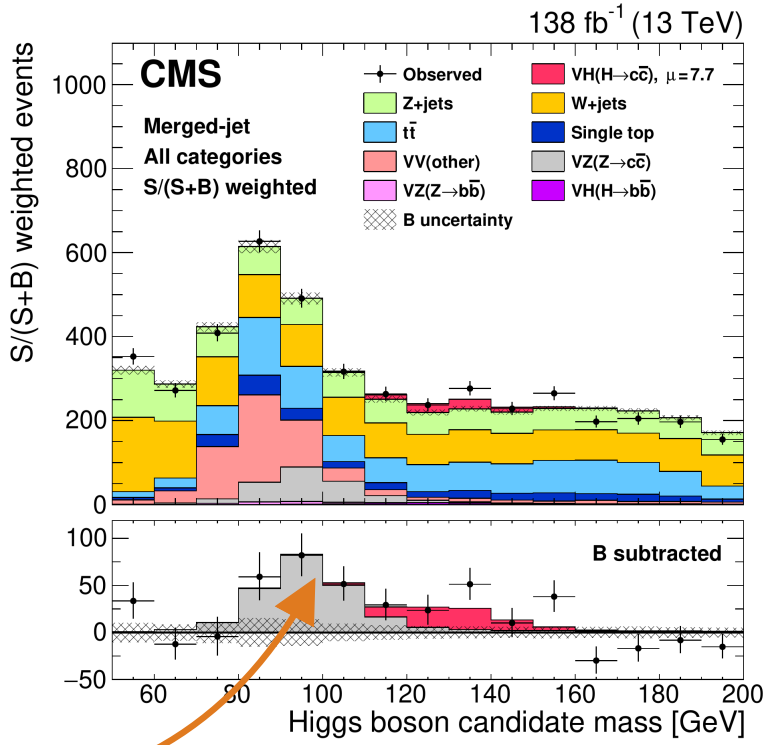
**Fit: BDT score (m($H_{cand}$) among the inputs)**

### Merged · one large-R jet (R = 1.5)

- **<5% of the signal**, but much less background
- **ParticleNet** cc̄ tagger (graph NN, mass-decorrelated): **4–7×** better rejection than DeepAK15
- ParticleNet mass regression: **≈50% better** mass resolution

**Fit: m($H_{cand}$) (BDTs decorrelated from it)**

Merged-jet candidate mass; bottom panel after background subtraction: VZ (grey, ≈ 90 GeV) and VH (red)

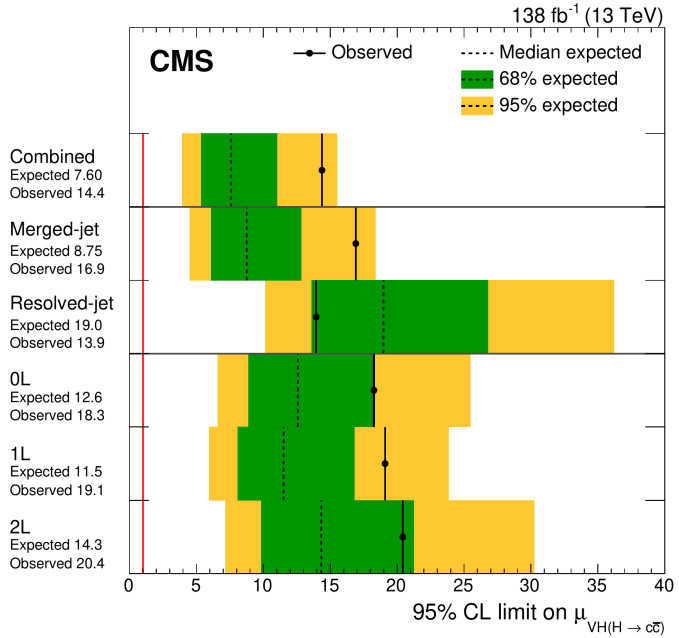



**Validation:** the same analysis sees **VZ(Z → cc̄) at 5.7σ** (grey peak, right): c tagging and background model work → backup

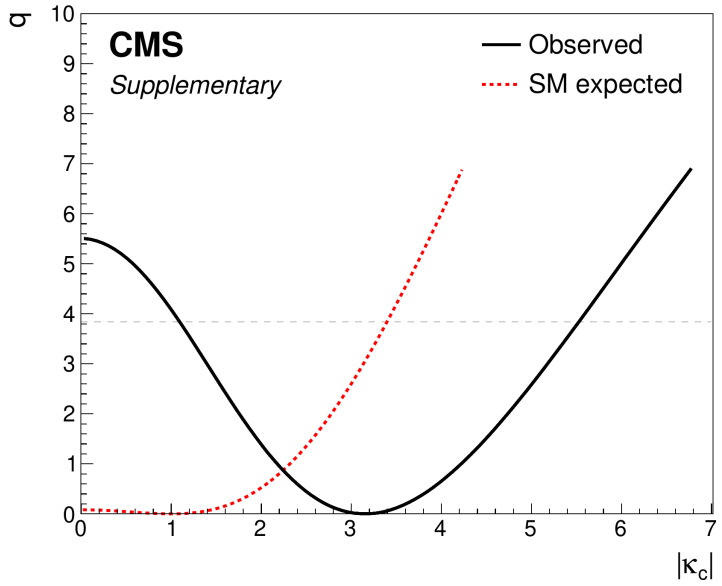
# VH(H → $c\bar{c}$): results

PRL 131 (2023) 061801 · CMS-HIG-21-008

## The most sensitive single channel of Run 2




95% CL limits on μ per category · red line: SM (μ = 1) · the **merged-jet** category carries most of the sensitivity




q = −2Δln L vs $|\kappa_c|$ (supplementary) · 95% CL where q < 3.84 (dashed line)

**85%**
**of the uncertainty is statistical**

**Statistics-limited: more data helps directly**

| | |
|---|---|
| Statistical | 85% |
| Experimental | 48% |
| Theory | 22% |

share of the total uncertainty on μ, in quadrature · details → backup

95% CL · observed (expected)

**μ < 14** (7.6) **1.1 < $|\kappa_c|$ < 5.5** (< 3.4)

μ limits with CLs; $\kappa_c$ interval from the profile-likelihood scan

***➜ Expected μ < 7.6: the best single channel of Run 2***

# Boosted H → $c\bar{c}$

PRL 131 (2023) 041801 · CMS-HIG-21-012

## Gluon-fusion enriched: one large-radius jet at high $p_T$

### ① Boosted regime: $p_T^H$ > 450 GeV

- High $p_T$ suppresses the QCD background
- H → $c\bar{c}$ in **one large-radius jet** (R = 0.8); soft drop grooms it and sharpens the mass $m_{SD}$
- Signal: 55% ggF, 25% VBF, 20% VH

### ② DeepDoubleX tagger

- Convolutional + recurrent network on tracks and secondary vertices
- **Passing** region (signal-rich) and **failing** region (background-rich)
- Mass-decorrelated: QCD in the passing region is **predicted from data** in the failing region

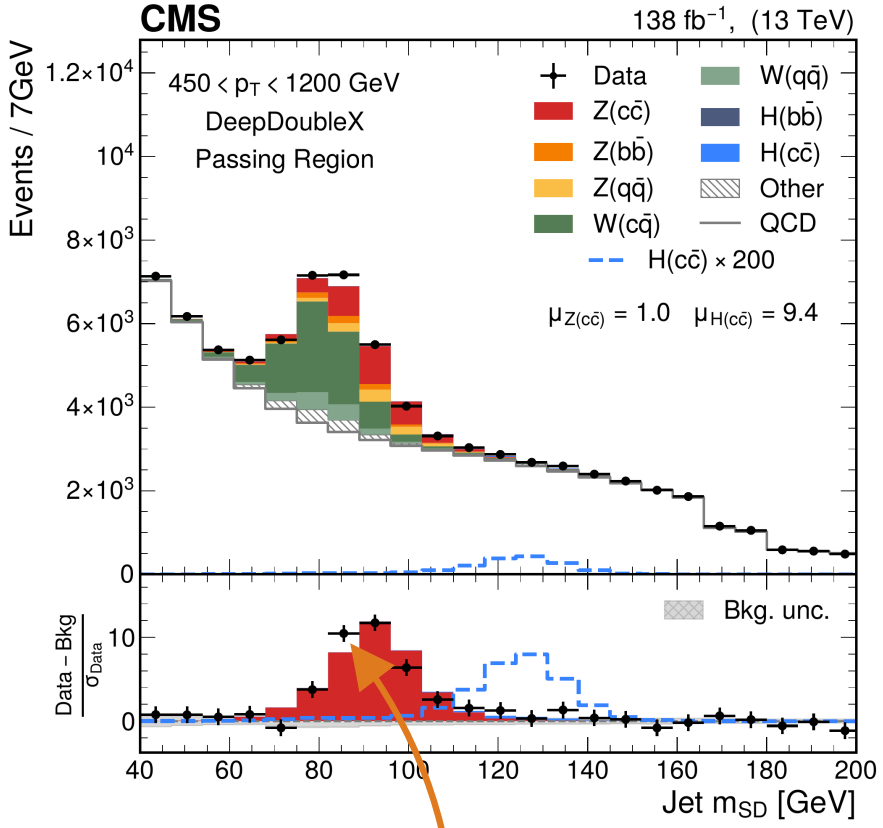
### ③ Statistical analysis

- Simultaneous binned fit of $m_{SD}$ in both regions, in 6 $p_T$ bins
- Upper limit from the profile likelihood ratio (CLs)

$$\sigma\mathcal{B} / (\sigma\mathcal{B})_{SM} < 47\ (39)$$

95% CL · observed (expected) · no $\kappa_c$ interpretation in the paper

Jet mass in the passing region · H → $c\bar{c}$ dashed (×200)




*Z → $c\bar{c}$ observed for the first time in Z+jets, well over 5σ*

# $t\bar{t}H(H \to c\bar{c})$: selection and jet tagging

PRL 136 (2026) 011801 · CMS-HIG-24-018

## Three $t\bar{t}$ decay channels, every jet flavour-tagged

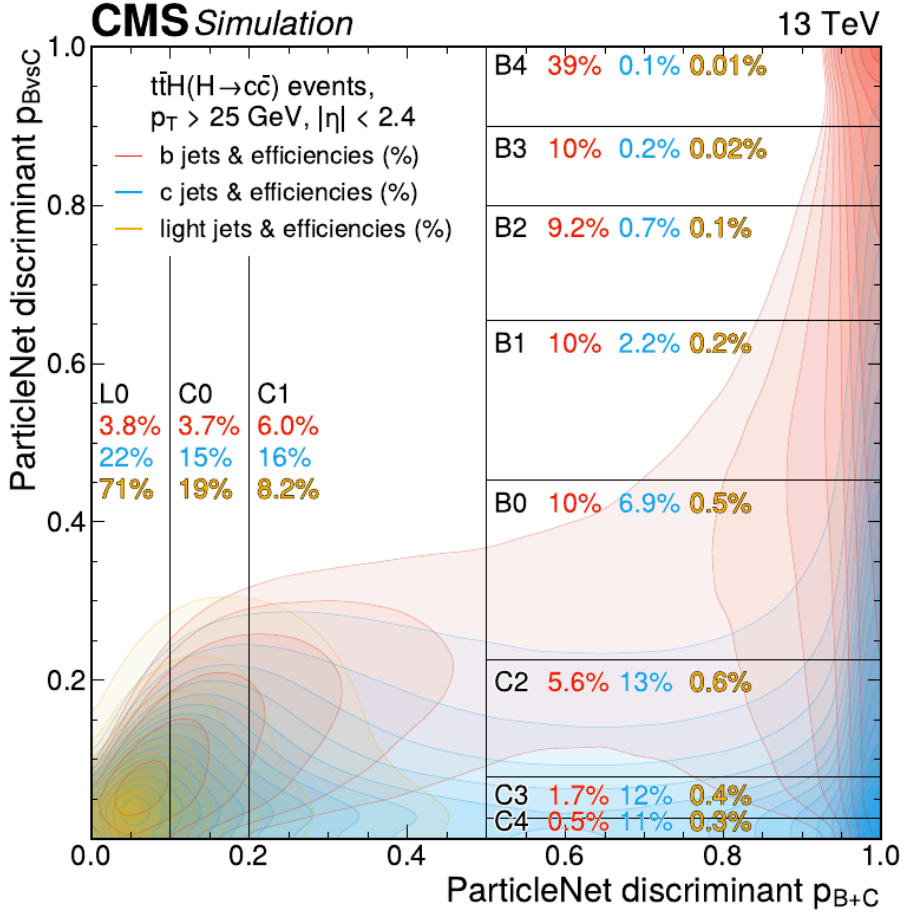
**0ℓ** — **≥ 7 jets**, $H_T$ > 500 GeV
fully hadronic $t\bar{t}$

**1ℓ** — **1 lepton + ≥ 5 jets**
single-lepton $t\bar{t}$

**2ℓ** — **2 opposite-charge leptons + ≥ 4 jets**, Z veto
dilepton $t\bar{t}$

**ParticleNet** (graph NN) tags every jet as b, c or light: **11 categories** (right), up to **2×** better background rejection than DeepJet

**Main background:** $t\bar{t}$+jets, split by the flavour of the extra jets: $t\bar{t}$ +b, +≥2b, +c, +≥2c, +light




b, c and light jets in the ParticleNet (b+c vs light, b vs c) plane, with the 11 tagging categories

# $t\bar{t}H(H \to c\bar{c})$: event classifier and regions

PRL 136 (2026) 011801 · CMS-HIG-24-018

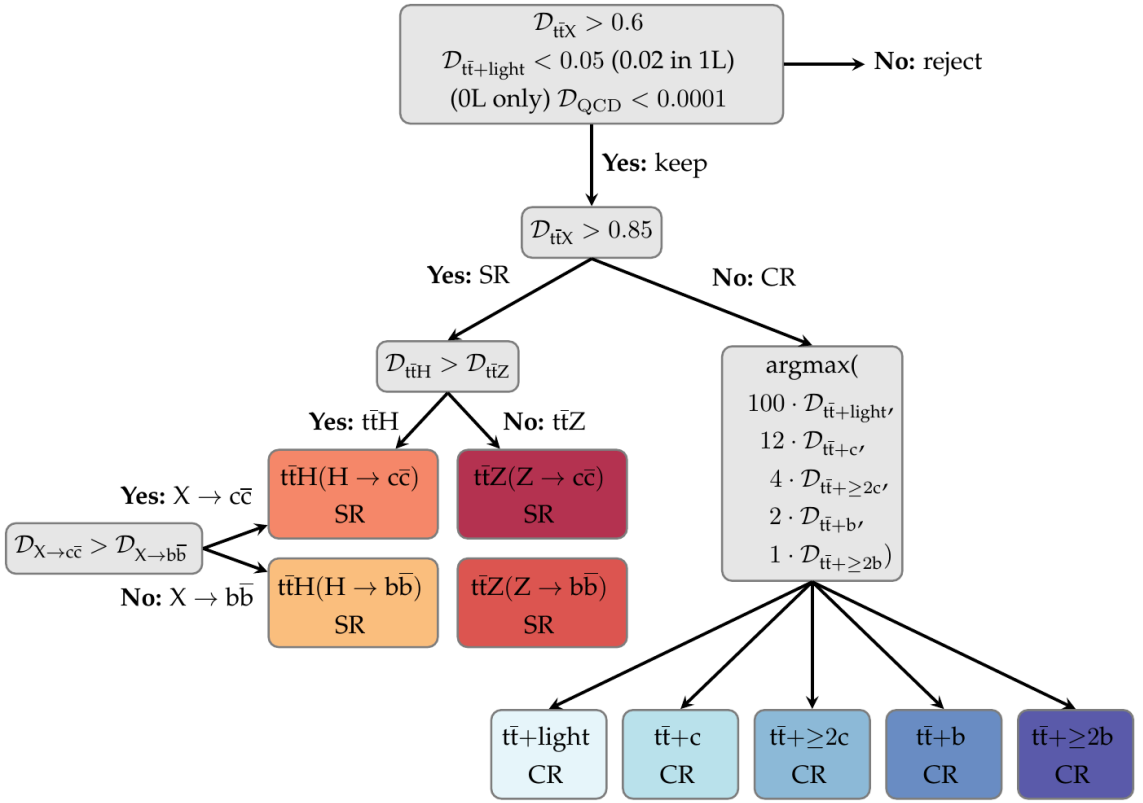
## Signal, control and validation regions from one classifier

**Event classifier: ParT** (transformer) on jets, leptons and $p_T^{miss}$: one score per process (4 signals, 5 $t\bar{t}$+jets, + QCD in 0ℓ)

$D_{t\bar{t}X}$ = sum of the four signal scores

| rejected | Validation | 5 control regions | 4 SRs |
|---|---|---|---|
| < 0.4 | 0.4–0.6 | 0.6–0.85 | > 0.85 |

- **SRs**: $t\bar{t}H$ / $t\bar{t}Z$ × $c\bar{c}$ / $b\bar{b}$; c-enriched SRs need ≥ 2 c jets
- **CRs**: one per $t\bar{t}$+jets flavour; their normalisations float in the fit
- **Validation**: same structure, no signal: good agreement → backup
- **QCD** (0ℓ): a classifier cut leaves < 1% of $t\bar{t}$+jets
- **Fit**: ParT score in all regions of the 3 channels at once

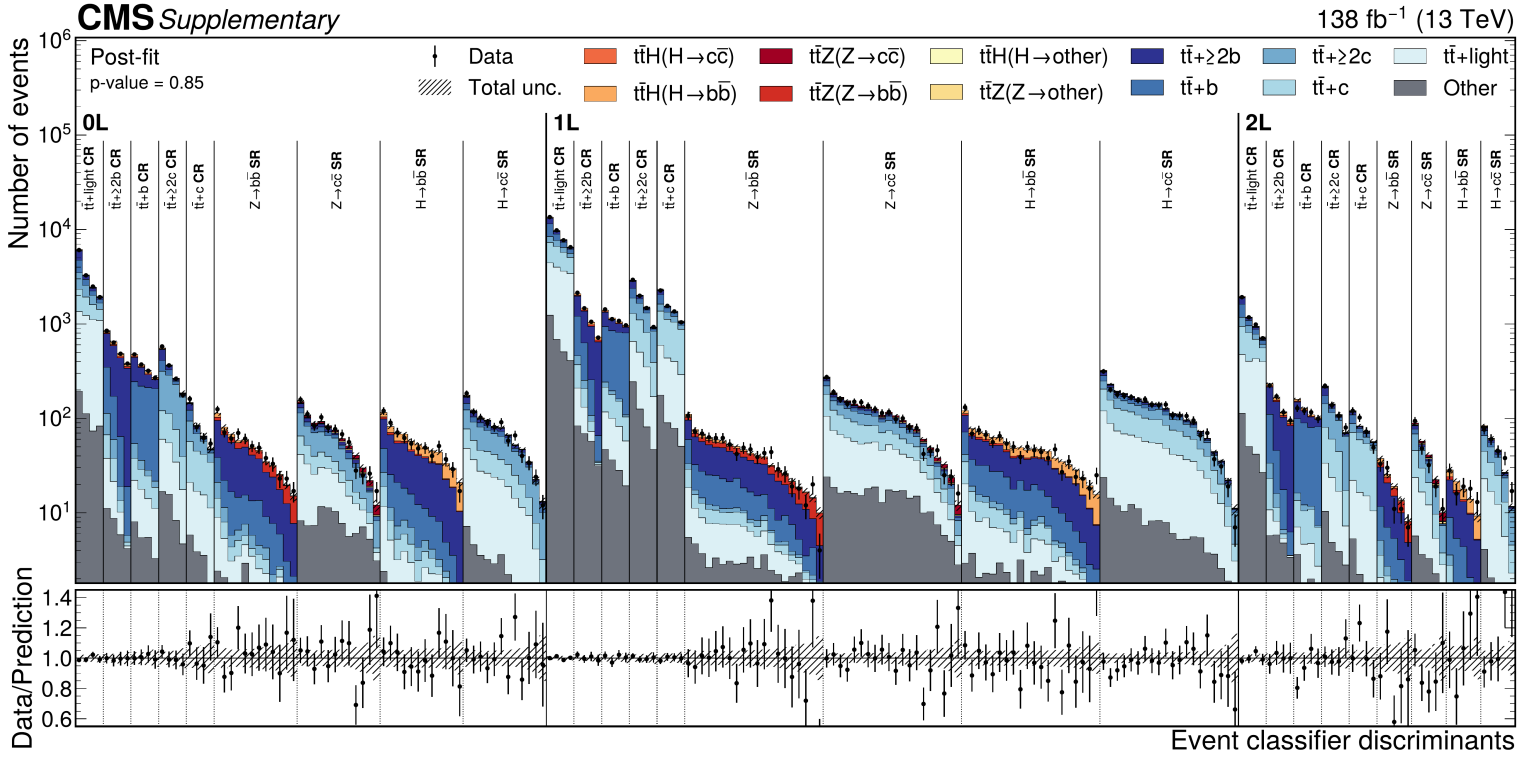



Categorisation of events with $D_{t\bar{t}X} > 0.6$ (same tree in the VR, 0.4–0.6)

# $t\bar{t}H(H \to c\bar{c})$: results

PRL 136 (2026) 011801 · CMS-HIG-24-018

## One fit to all regions: four signals and every $t\bar{t}$+jets component




Post-fit ParT discriminants in the 5 CRs and 4 SRs of each channel (0ℓ, 1ℓ, 2ℓ) · fit p-value 0.85

$t\bar{t}H(H \to c\bar{c})$ · 95% CL · observed (expected)

**$\mu < 7.8$ (8.7)**

**$|\kappa_c| < 3.0$ (3.3)**

$\kappa_b = 1$ · with $\kappa_b$ fitted: $|\kappa_c| < 4.2$ obs. → backup

| | best fit $\hat{\mu}$ | obs. (exp.) |
|---|---|---|
| $t\bar{t}H(b\bar{b})$ | $0.91\ ^{+0.26}_{-0.22}$ | 4.4σ (4.5σ) |
| **$t\bar{t}H(c\bar{c})$** | **−1.6 ± 4.5** | – |
| $t\bar{t}Z(b\bar{b})$ | $1.47\ ^{+0.45}_{-0.41}$ | 3.5σ (2.4σ) |
| $t\bar{t}Z(c\bar{c})$ | $1.02\ ^{+0.79}_{-0.84}$ | 1.2σ (1.3σ) |

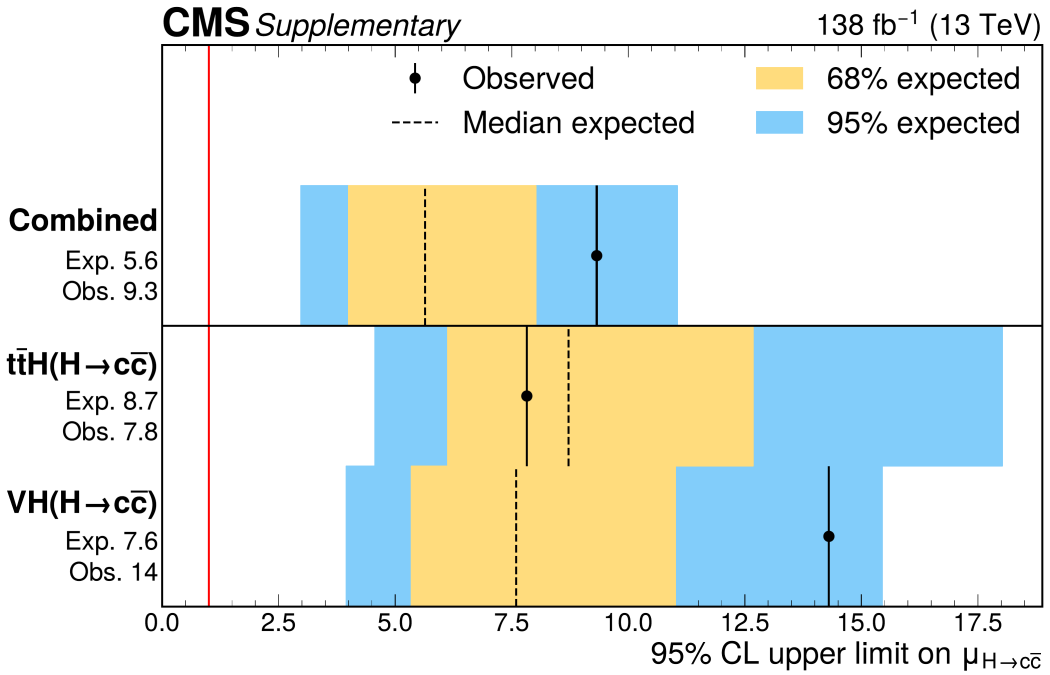
$t\bar{t}Z(c\bar{c})$ and $t\bar{t}Z(b\bar{b})$ are the standard candles: both consistent with the SM · limits per channel and event yields → backup

***→ A second direct channel, with expected sensitivity comparable to VH***

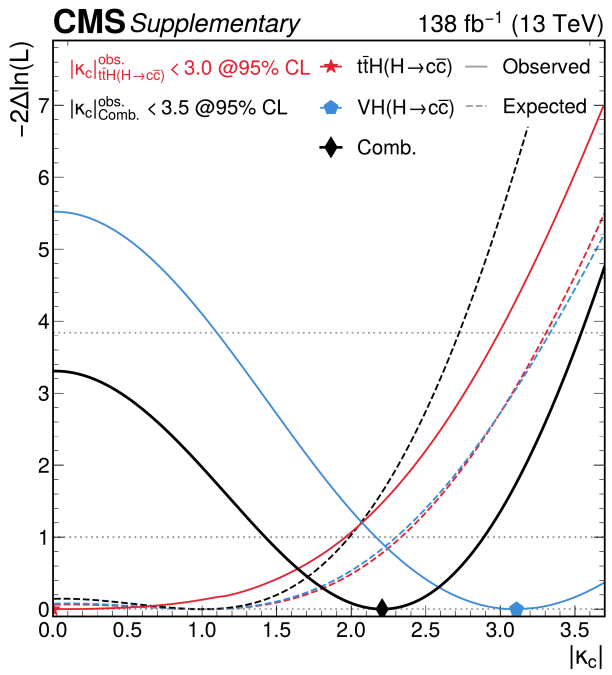
# $t\bar{t}H$ + VH combination

PRL 136 (2026) 011801 · CMS-HIG-24-018

## Best expected direct sensitivity to $\kappa_c$




95% CL limits on $\mu_{H\to c\bar{c}}$: $t\bar{t}H$, VH and combined




$-2\Delta\ln L$ vs $|\kappa_c|$ ($\kappa_b = 1$)

$t\bar{t}H$ + VH · 95% CL
observed (expected)

**$\mu < 9.3$ (5.6)**

**$|\kappa_c| < 3.5$ (2.7)**

SM production, $\kappa_b = 1$

**26%**
better expected μ limit than VH alone

**36%**
better expected μ limit than $t\bar{t}H$ alone

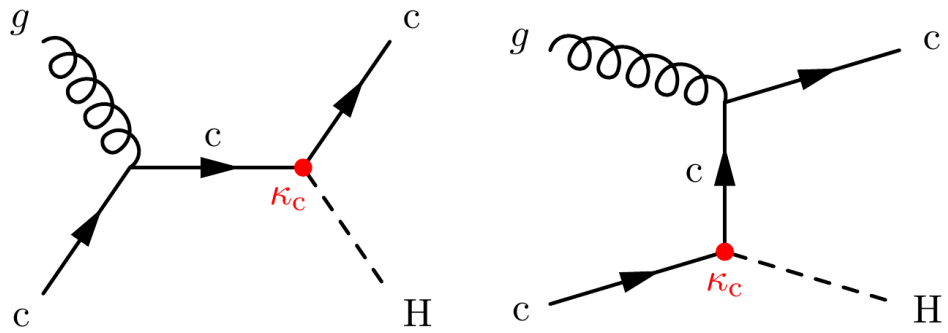
**$|\kappa_c| < 2.7$**
expected, below the BSM-motivated ~3

***➜ The two direct channels together set the best expected constraint on $\kappa_c$***

# H + charm production

JHEP 11 (2025) 060 · JHEP 08 (2026) 051

## $\kappa_c$ in the production vertex, independent of the H → $c\bar{c}$ decay




LO diagrams for cH production; $\kappa_c$ at the marked vertices

**Signal**: a clean Higgs decay (γγ or WW) + one c-tagged jet

**"Flat direction"**: $\kappa_c$ varies, all other Higgs rates stay SM, so only cH production changes

## Challenges

### 1 Tiny rate

$\sigma_{cH}$ = 90 fb, about 0.2% of all Higgs production: $\sigma\mathcal{B}$ ≈ **0.2 fb** (γγ), ≈ 20 fb (WW)

### 2 Yukawa-insensitive processes

ggH and other Higgs processes with an extra c jet (e.g. from g → $c\bar{c}$) give the **same final state** without involving $\kappa_c$; treated as background

### 3 b-associated Higgs (bH)

$\sigma_{bH}$ = 660 fb, 7× larger: b jets mistagged as c, so c-vs-b separation matters

### 4 Modelling

Flavour scheme for cH (4FS vs 3FS): **25–30%** uncertainty; heavy-flavour modelling of ggH (T. Bevilacqua, next talk)

***→ A complementary handle on $\kappa_c$, but small rates and irreducible H+c make it hard***

JHEP 11 (2025) 060 · CMS-HIG-23-010

# cH(H → γγ)

## First search for H + charm in the diphoton channel

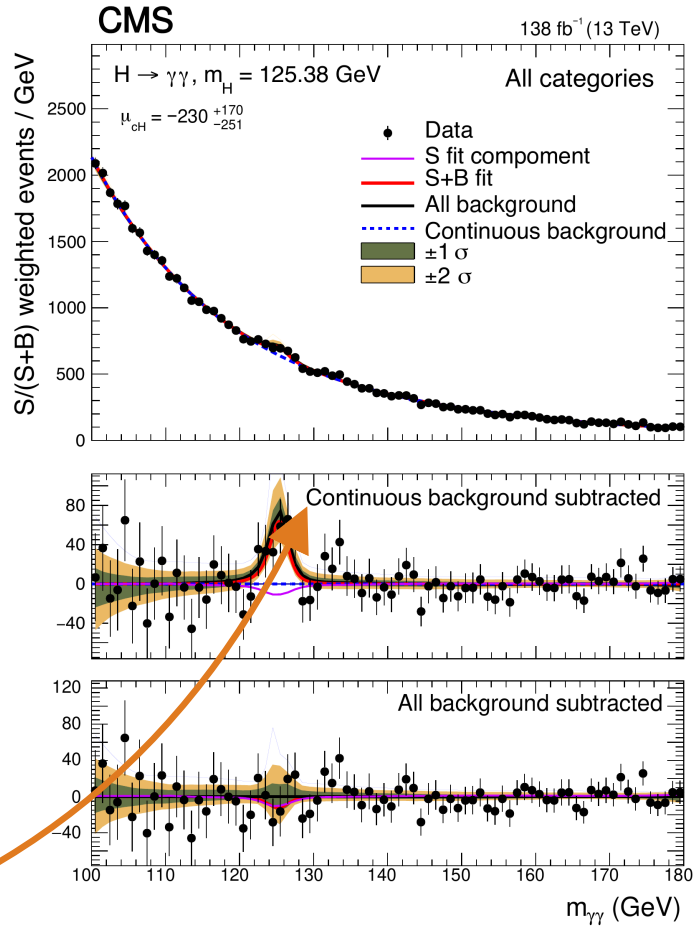
- 138 $fb^{-1}$, DeepJet c tagging
- Two BDTs, without c-tag inputs: cH vs ggH (largest resonant background) and cH vs continuum γγ+jets
- 27 categories (9 per year); signal from fits to $m_{\gamma\gamma}$

**$\mu_{cH} < 243$ (355)** **$|\kappa_c| < 38$ (73)**

95% CL · observed (expected)

*$|\kappa_c| \lesssim 40$, not ~3: $\sigma\mathcal{B} \approx 0.2$ fb, and ggH H+c mimics the signal*

*125 GeV peak is fully explained by the other Higgs processes (mostly ggH): $\hat{\mu}_{cH} = -230^{+170}_{-251}$*




$m_{\gamma\gamma}$, all categories, S/(S+B) weighted

# cH(H → WW) and the γγ + WW combination

JHEP 08 (2026) 051 · CMS-HIG-24-009

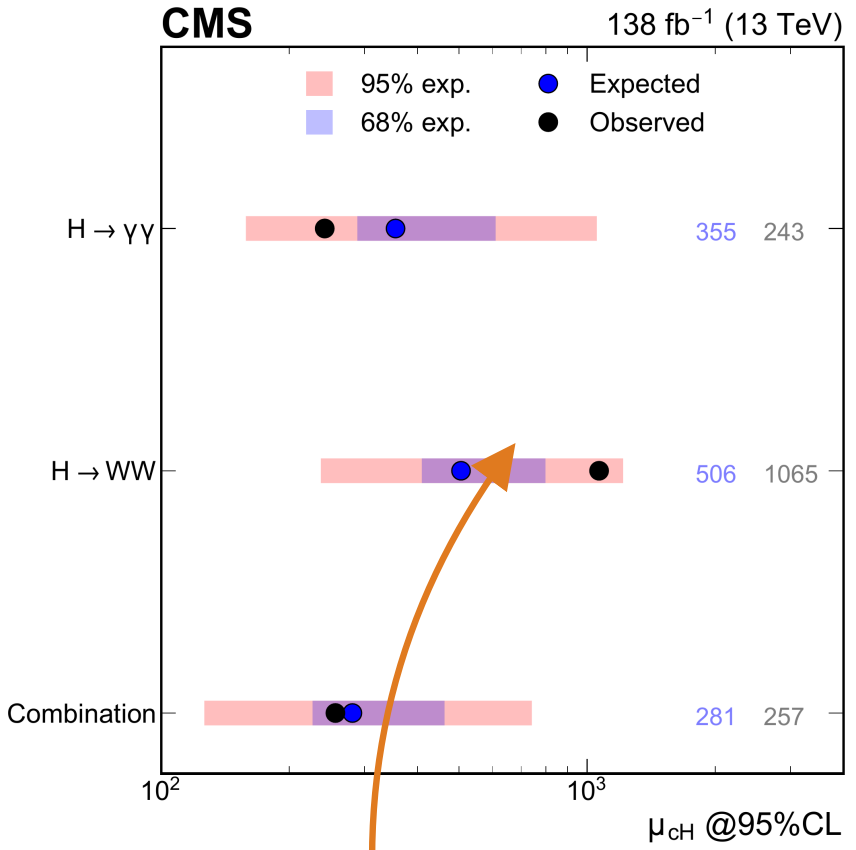
## eνμν + a c jet, then combined with γγ

- Signal regions with 1 or >1 DeepJet c jets; two BDTs: vs non-Higgs and vs other Higgs processes
- $\kappa_c$-independent H+c fixed to the SM: $\mu_{cH}$ is the only parameter of interest (floating it → backup)
- WW alone: $\mu_{cH} < 1065$ (506), $|\kappa_c| < 211$ (95)

γγ + WW · 95% CL · observed (expected)

**$\mu_{cH} < 257$ (281)** **$|\kappa_c| < 47$ (51)**

*$\kappa_c$ from production alone: statistics-limited*




$\mu_{cH}$ limits: γγ, WW, combined

*WW fluctuated up ($\hat{\mu}_{cH}$ = 390 vs −230 in γγ): obs. weaker than γγ alone (38)*

# Indirect constraints on $\kappa_c$

PRD 112 (2025) 112001 · arXiv:2607.23352 · PLB 865 (2025) 139462

## The Higgs width from on- and off-shell H → ZZ → 4ℓ

$$\sigma_{on} \propto \frac{g_p^2\, g_d^2}{\Gamma_H} \qquad \sigma_{off} \propto g_p^2\, g_d^2$$

$g_p$, $g_d$: production and decay couplings ($g_d \propto \kappa_Z$). Large $\kappa_c$ adds $c\bar{c} \to H$ and raises $\Gamma_H$. On-shell alone, a larger $\kappa_Z$ can hide it, so $|\kappa_Z| \leq 1$ must be assumed. **Off-shell has no $\Gamma_H$**: it fixes $\kappa_Z$ and removes the assumption.

| $\kappa_c$ at 95% CL | observed | expected |
|---|---|---|
| On-shell, $\lvert\kappa_Z = \kappa_W\rvert \leq 1$ | [−3.7, 3.2] | [−3.8, 3.3] |
| **new** On + off-shell, $\lvert\kappa_V\rvert \leq 1$ | **[−3.3, 2.8]** | [−3.5, 2.9] |
| **new** On + off-shell, $\kappa_Z$, $\kappa_W$ free | [−8.6, 7.5] | [−9.6, 8.3] |

H → J/ψ γ: $-166 < \kappa_c < 208$ (J. Bierkens, Thu) · Higgs width: F. Errico (Tue) · $p_T^H$ spectrum → backup

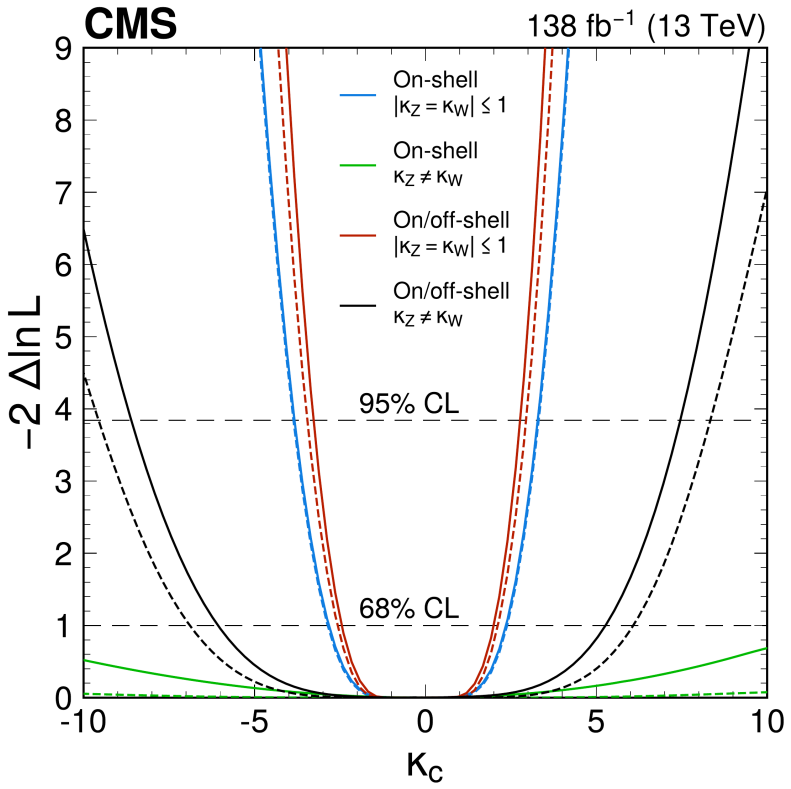
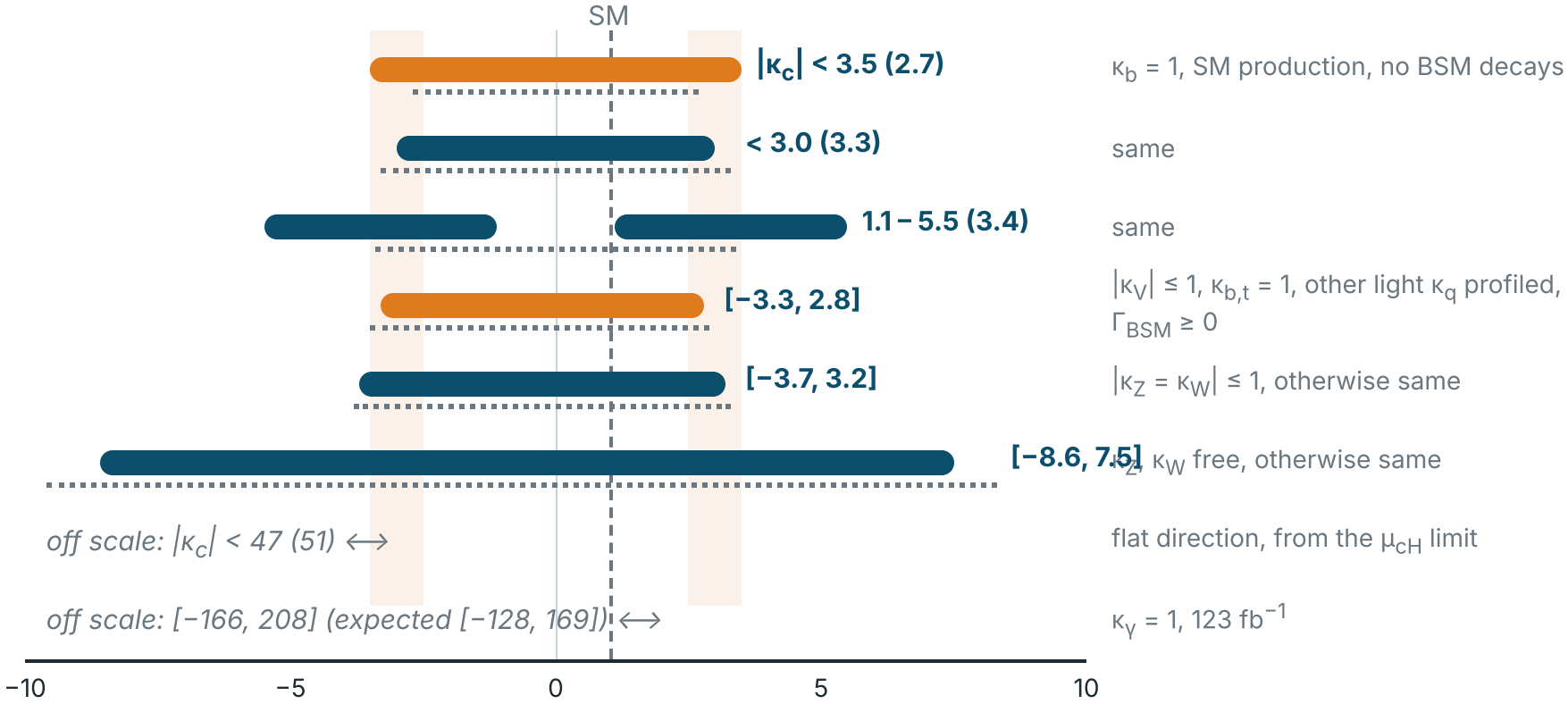



Fig. 8: on-shell (blue, green), on+off-shell (red, black), with $|\kappa_Z = \kappa_W| \leq 1$ or $\kappa_Z \neq \kappa_W$; solid obs., dashed exp. Table: Table 3, no custodial symmetry

***→ With $|\kappa_V| \leq 1$, H → 4ℓ matches the direct searches, under different assumptions***

# Where we stand

## CMS constraints on $\kappa_c$ at 95% CL




**$t\bar{t}H + VH(H \to c\bar{c})$**
PRL 136 (2026) 011801

**$t\bar{t}H(H \to c\bar{c})$**
PRL 136 (2026) 011801

**$VH(H \to c\bar{c})$**
PRL 131 (2023) 061801

**On + off-shell H → 4ℓ**
arXiv:2607.23352

**On-shell H → 4ℓ**
PRD 112 (2025) 112001

**On + off-shell H → 4ℓ**
arXiv:2607.23352

**cH (γγ + WW)**
JHEP 08 (2026) 051

**H → J/ψ γ**
PLB 865 (2025) 139462

bar: observed · dotted: expected · shaded: $|\kappa_c| \sim 3$ (BSM) · dashed: SM · the rows use different coupling models and are not directly comparable

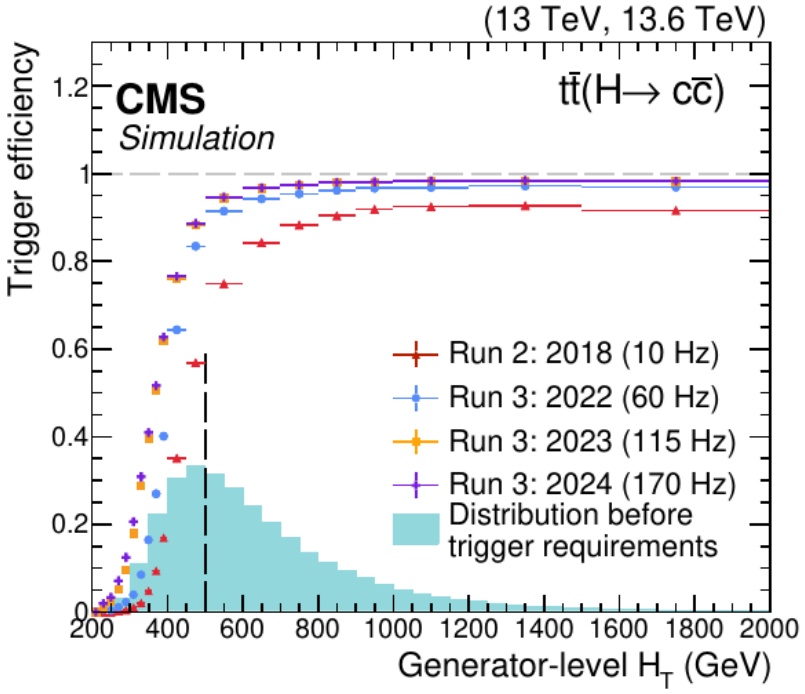
$t\bar{t}H$ alone beats the combination on observed because VH fluctuated up; the combination has the best *expected* sensitivity (2.7)

***→ Direct searches and on + off-shell H → 4ℓ both reach $|\kappa_c| \sim 3$, where flavour-compatible BSM lives***

# Toward Run 3

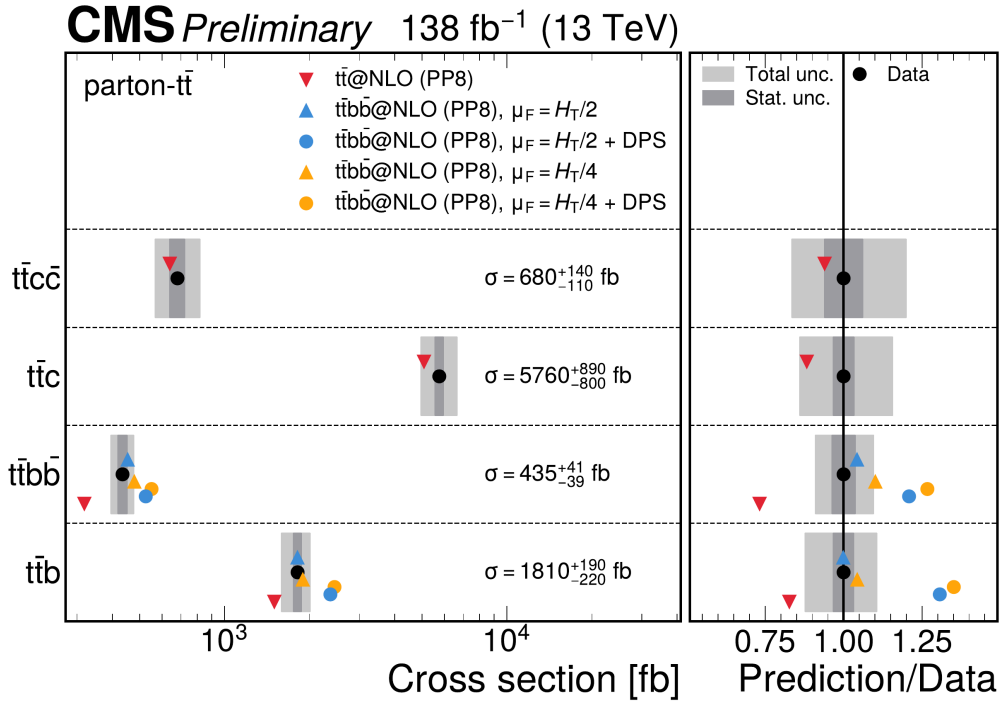
## Better triggers, and the main $t\bar{t}H$ background measured

### Run 3: ML b/c tagging at the HLT




arXiv:2608.18331 (BTV-25-002), simulation: $t\bar{t}H(H \to c\bar{c})$ trigger efficiency, Run 2 (2018) vs Run 3 (2022–24), 13.6 TeV · HLT taggers: J. Daniel (Mon) → backup

### Run 2: $t\bar{t}c\bar{c}$ cross section measured




CMS-PAS-TOP-26-003 **new**, 138 fb$^{-1}$: $\sigma_{t\bar{t}c\bar{c}}$ = 680 ± 40 (stat) $^{+130}_{-100}$ (syst) fb

***➜ Run 3 starts with higher trigger efficiency and a $t\bar{t}$+$c\bar{c}$ background measured in Run 2***

# Run 3 and HL-LHC prospects

## From $|\kappa_c| \sim 3$ toward $\kappa_c$ of order one

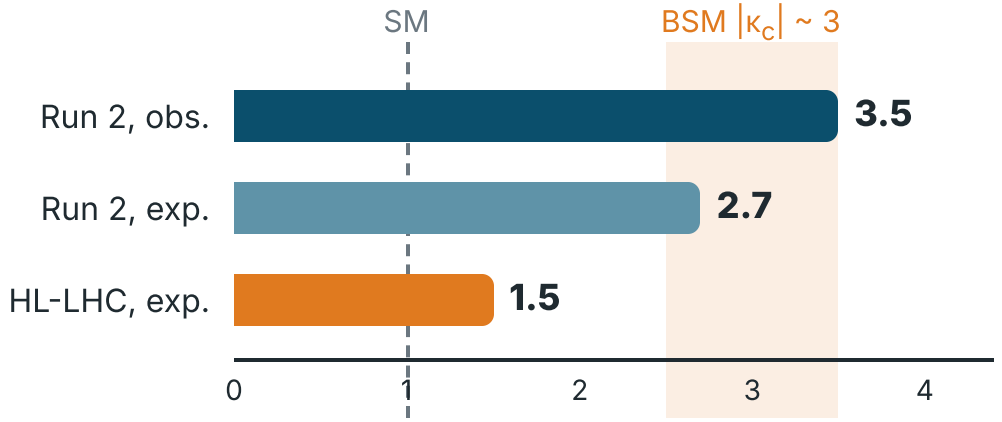
### Run 3 (13.6 TeV)

- The direct searches are **statistics-limited** (VH: 85% of the uncertainty): more data helps directly
- UParT rejects more b and light jets than DeepJet and ParticleNet at the same c efficiency (slide 3)
- HLT b/c tagging and a measured $t\bar{t}c\bar{c}$ background (slide 17)
- ATLAS VBF H → $c\bar{c}$ already uses 51.5 $fb^{-1}$ of Run 3 data

### HL-LHC (3 $ab^{-1}$ per experiment)

- ATLAS + CMS European Strategy input (arXiv:2504.00672, CMS-HIG-25-002): VH(H → $c\bar{c}$) ≈ 1.6σ expected
- **Expected $|\kappa_c| < 1.5$** (95% CL, 14 TeV): VH with improved c tagging (ATLAS study, assumed for both experiments); 30% better than the 2019 Yellow Report
- Next gains: Run 3 data, ML tagging, detector upgrades; theory modelling and tagger calibration become limiting

### Direct searches: allowed $|\kappa_c|$




Run 2: $t\bar{t}H$ + VH (PRL 136 (2026) 011801) · HL-LHC: ATLAS + CMS, 3 $ab^{-1}$ each · every bar still includes $\kappa_c = 0$

$\kappa_c$ of order one: first sensitivity to whether the Higgs gives charm its mass

# Summary

## Does the Higgs give charm its mass?

The charm Yukawa is the most accessible second-generation quark coupling; CMS probes it **in the decay, in production and indirectly**

### Direct

$t\bar{t}H$ + VH($H \to c\bar{c}$)

**$|\kappa_c| < 3.5$** (2.7)

most stringent direct constraint, best expected sensitivity

### Production

H + charm, γγ + WW

**$|\kappa_c| < 47$** (51)

independent of the $H \to c\bar{c}$ decay

### Indirect

on + off-shell $H \to 4\ell$

**$[-3.3, 2.8]$**

with $|\kappa_V| \leq 1$ (about ±8 without)

### Next

Run 3 data

**better taggers and triggers**

$t\bar{t}c\bar{c}$ background already measured

Today **$|\kappa_c| \lesssim 3$–$3.5$**: entering BSM-motivated territory

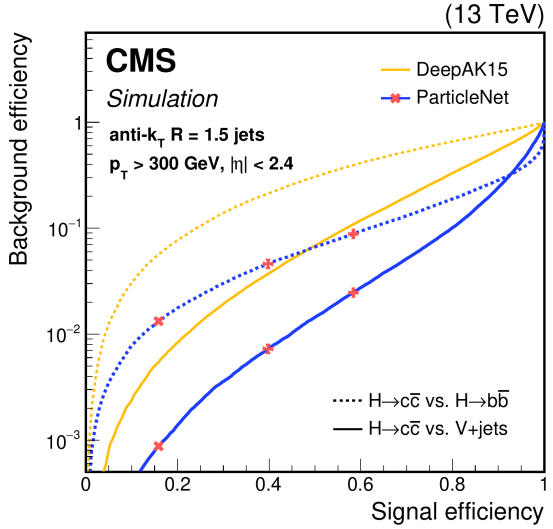
HL-LHC expected **$|\kappa_c| < 1.5$**: approaching, not yet reaching, sensitivity to $\kappa_c = 0$

***➜ Thank you!***

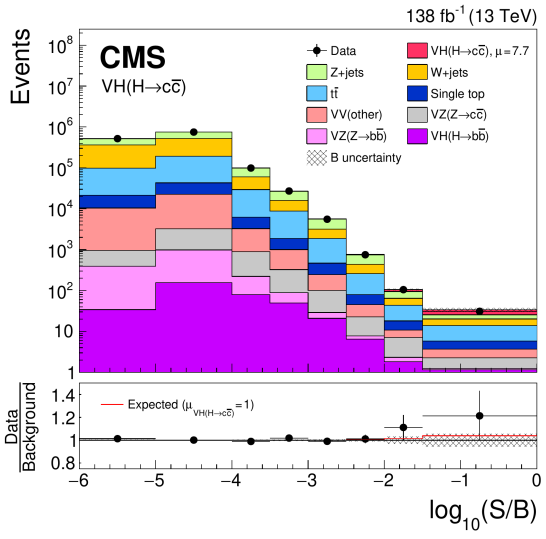
# Backup

# VH(H → $c\bar{c}$): categories and sensitivity

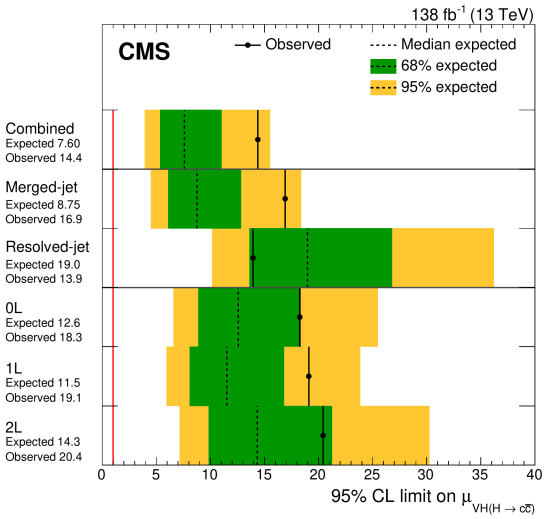
PRL 131 (2023) 061801



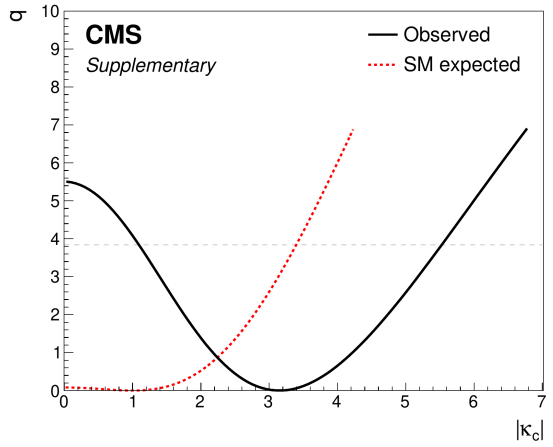

ParticleNet merged-jet H → $c\bar{c}$ tagging




Events vs $\log_{10}$(S/B)




μ limits per channel (inner band 68%)




|$\kappa_c$| scan (supplementary)

Best fit μ = 7.7 $^{+3.8}_{-3.5}$; merged μ = 8.7 $^{+4.6}_{-4.0}$, resolved μ = −9.5 ± 9.6 · σ$\mathscr{B}$ < 0.94 (0.50) pb · μ limits with CLs, $\kappa_c$ intervals from profile-likelihood scans

➜ *The upward fluctuation comes from the merged-jet category*

# VH(H → $c\bar{c}$): analysis details

PRL 131 (2023) 061801 · strategy as in JHEP 03 (2020) 131

**Topologies** Two, orthogonal: H candidate from a large-R (R = 1.5) jet with $p_T$ above (merged) or below (resolved) 300 GeV; same lepton channels

**Resolved** DeepJet CvsL/CvsB; the two highest-CvsL jets form the H candidate; leading-jet cuts at ≈40% c efficiency (≈4% light, ≈16% b mistag). c-jet energy regression, FSR-jet recovery and a 2ℓ kinematic fit improve the $m(H_{cand})$ resolution by up to 20%. BDT per channel, fit to the BDT score

**Merged** ParticleNet $c\bar{c}$ tagger, 4–7× better rejection than DeepAK15; 3 working points (≈58, 40, 16% $c\bar{c}$ efficiency) define 3 purity categories. ParticleNet mass regression: ≈50% better mass resolution. BDTs decorrelated from $m(H_{cand})$; fit to $m(H_{cand})$

**Backgrounds** V+jets and $t\bar{t}$ normalised in dedicated control regions, following the 2016 analysis (V+jets split by jet flavour, free rate parameters)

**VZ validation** Same analysis, resolved BDTs retrained with VZ as signal, merged unchanged. $\mu_{VZ} = 1.01\ ^{+0.23}_{-0.21}$; 4.4σ merged, 3.1σ resolved, 5.7σ combined (exp. 4.7, 3.3, 5.9)

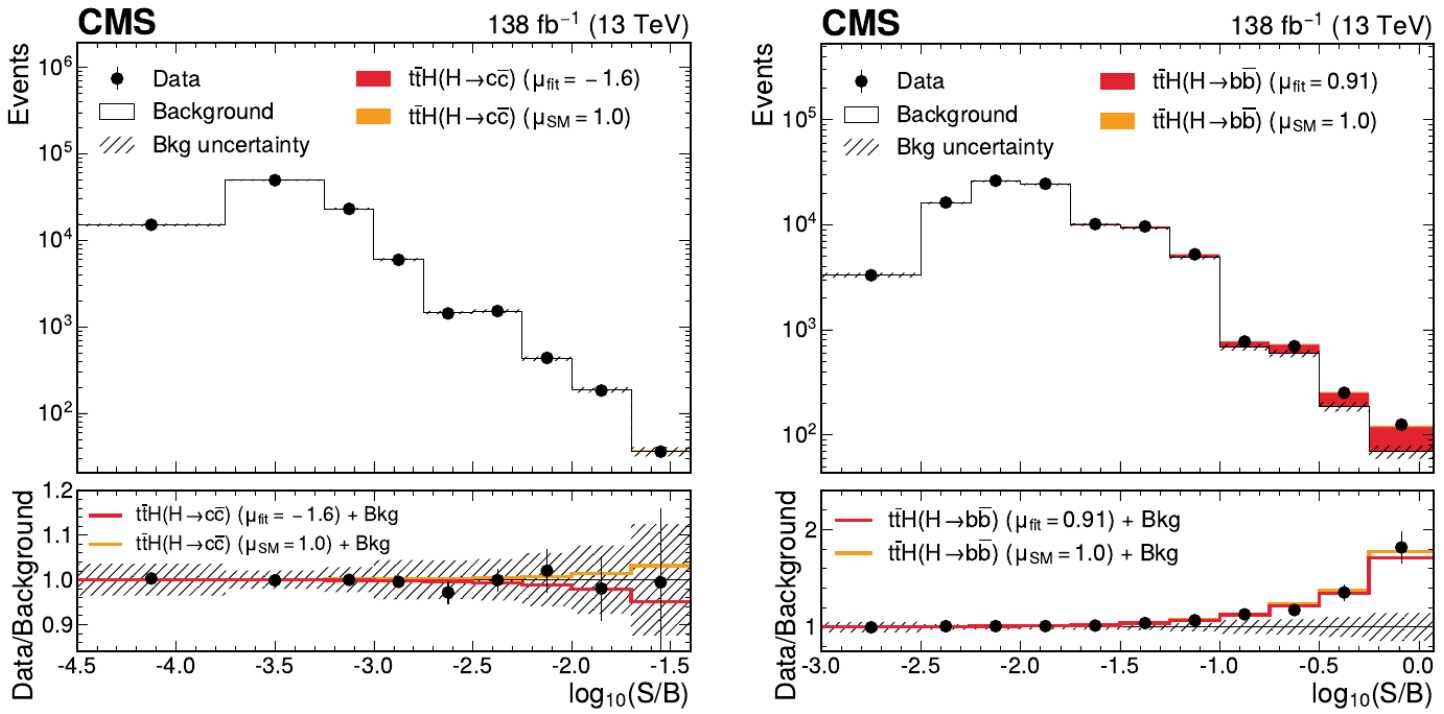
| Source | $\Delta\mu/\Delta\mu_{tot}$ |
|---|---|
| **Statistical** | **85%** |
| Bkg. normalisations | 37% |
| Experimental | 48% |
| Simulated sample sizes | 37% |
| c-jet identification | 23% |
| Jet energy scale, resolution | 15% |
| Simulation modelling | 11% |
| Luminosity | 6% |
| Lepton identification | 4% |
| Theory | 22% |
| Backgrounds | 17% |
| Signal | 15% |

For $\hat{\mu} = 7.7\ ^{+3.8}_{-3.5}$; simulation size driven by negative weights in NLO V+jets

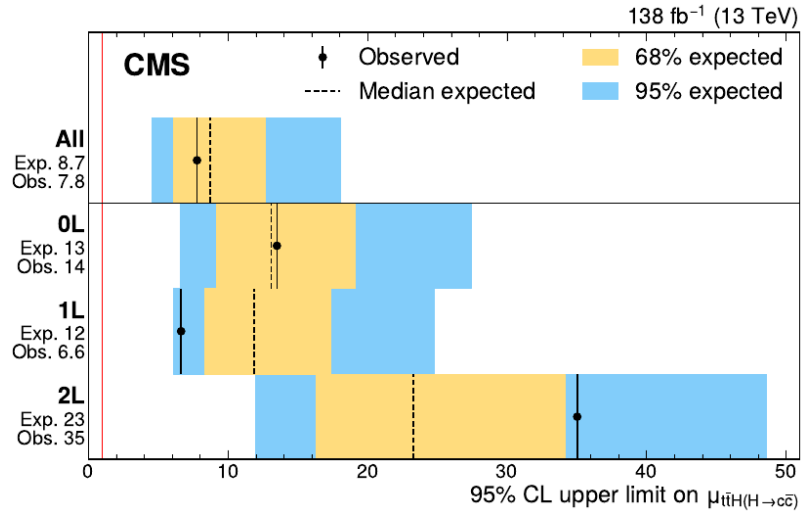
***➜ Statistics-limited; c tagging and simulation size come next***

# $t\bar{t}H(H \to c\bar{c})$: event yields

PRL 136 (2026) 011801




Event yields vs $\log_{10}(S/B)$; the $t\bar{t}H(c\bar{c})$ panel shows the published best fit




$\mu_{ttH(c\bar{c})}$ limits per channel (inner band 68%)

**→ *$t\bar{t}H(c\bar{c})$ best fit $\hat{\mu} = -1.6 \pm 4.5$: compatible with zero and with the SM***

# $t\bar{t}H(H \to c\bar{c})$: flavour tagging and by-products

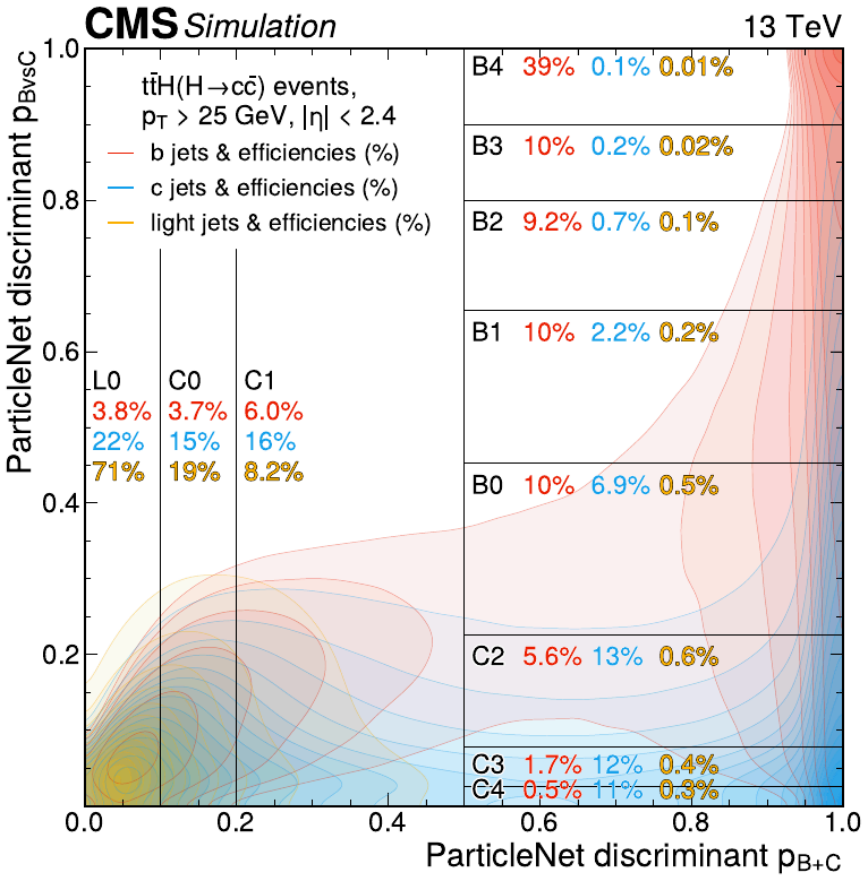
PRL 136 (2026) 011801

## Tagging

- Each jet gets ParticleNet scores; the (b+c vs light, b vs c) plane is split into 2D categories
- ParT event classifier separates $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z$, $t\bar{t}$ +HF

## By-products

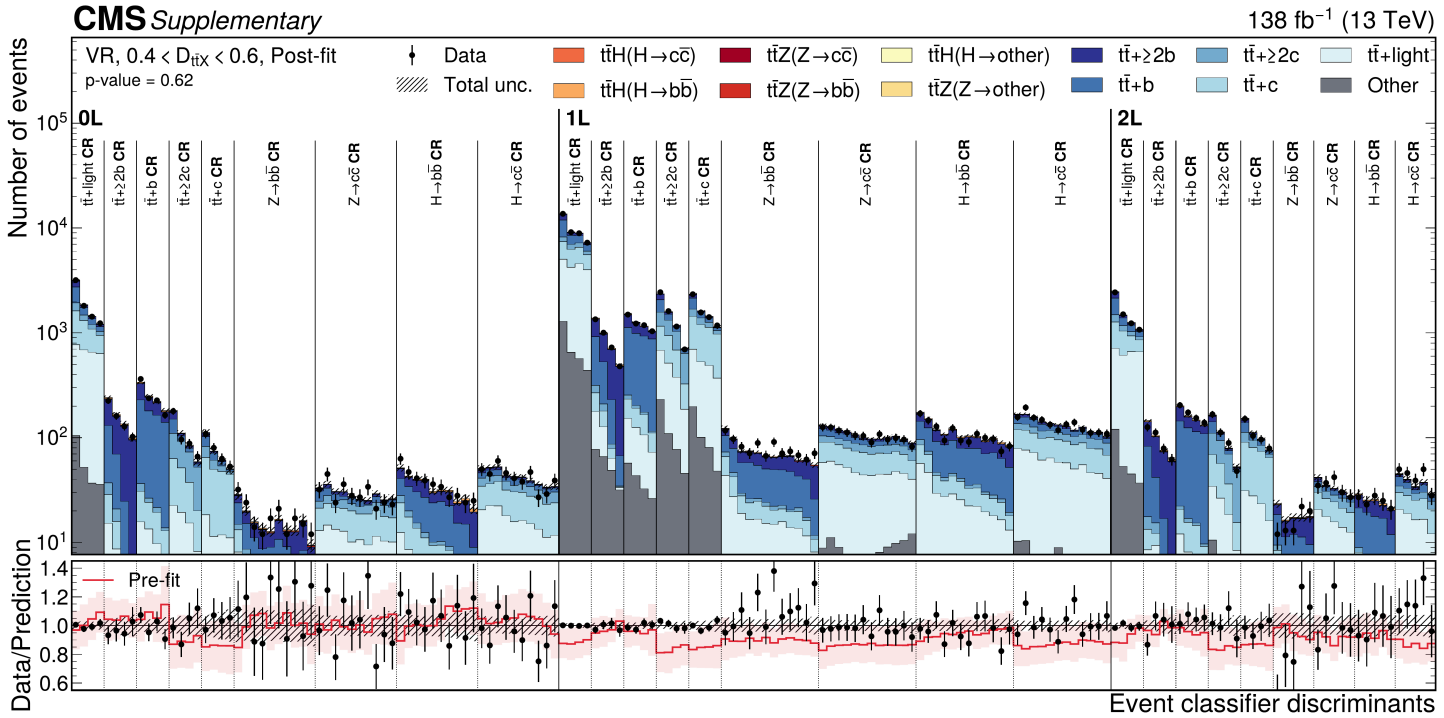
- $t\bar{t}Z(Z \to c\bar{c})$ standard candle: $\mu = 1.0 \pm 0.8$
- $t\bar{t}H(H \to b\bar{b})$: $\mu = 0.91\ ^{+0.26}_{-0.22}$, $4.4\sigma$ ($4.5\sigma$ exp.)




b, c, light jets in the ParticleNet plane and the 2D tagging categories

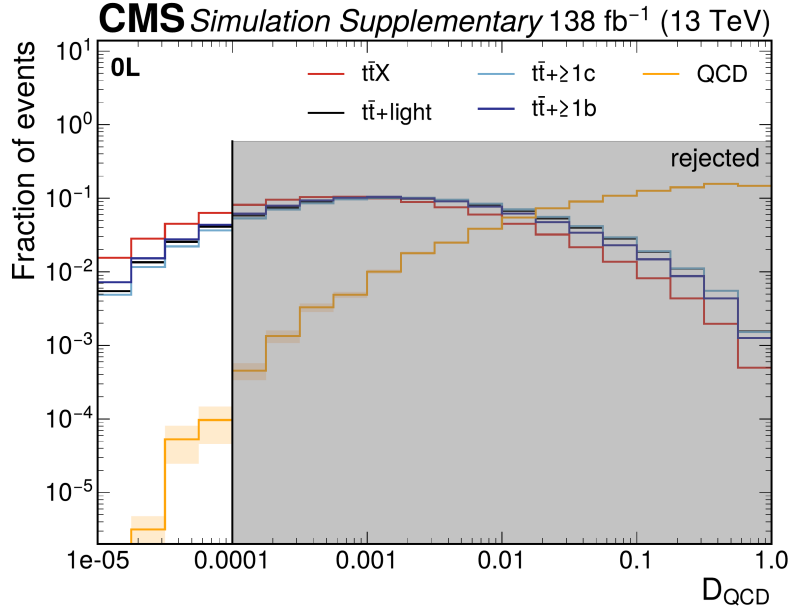
# $t\bar{t}H$: validation region and QCD rejection

PRL 136 (2026) 011801 (supplementary)




Validation region 0.4 < $D_{t\bar{t}X}$ < 0.6, same SR/CR structure (CR-like 0.40–0.58, SR-like 0.58–0.60): good agreement

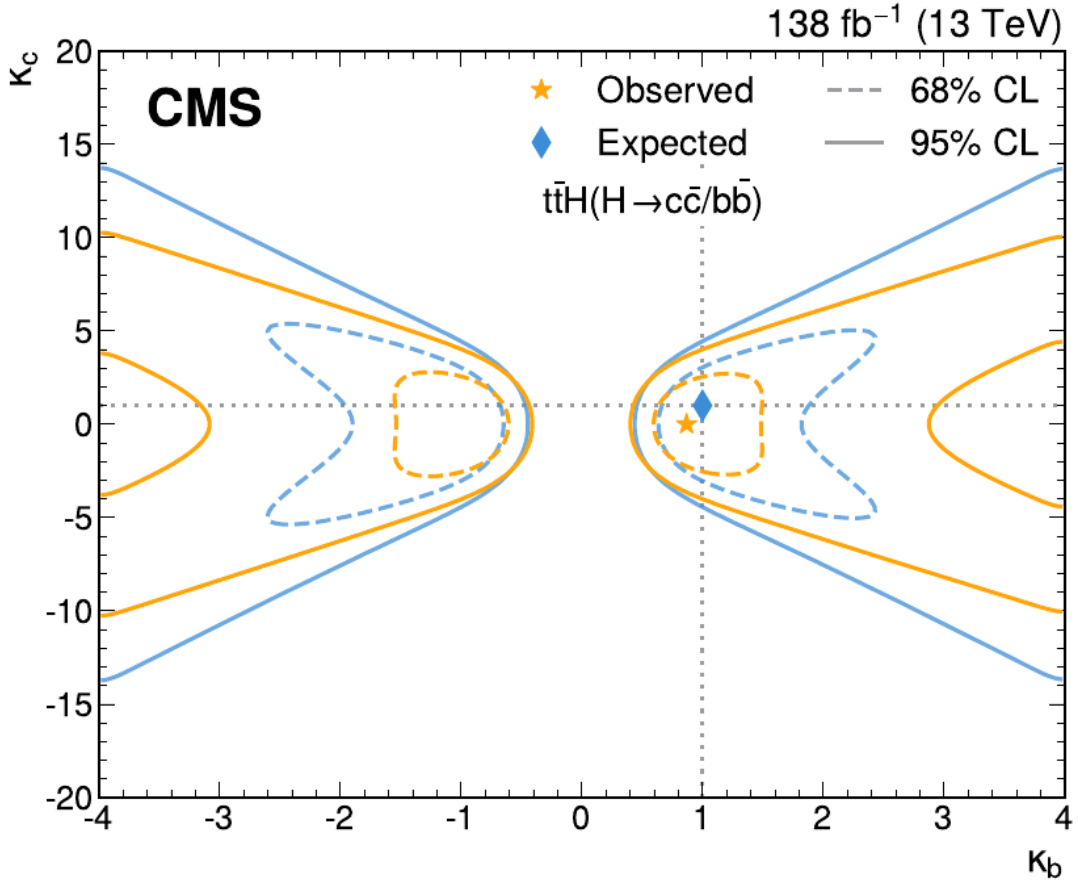
- c-enriched regions need ≥2 c jets ($t\bar{t}$+c: ≥1); CR choice weights 100, 12, 4, 2, 1 on the $t\bar{t}$+light, +c, +≥2c, +b, +≥2b scores
- $t\bar{t}$+b, +≥2b from a dedicated 4FS $t\bar{t}b\bar{b}$ sample




0ℓ only: $D_{QCD}$ < $10^{-4}$ (grey area rejected) suppresses QCD by nearly four orders of magnitude, to <1% of $t\bar{t}$+jets; QCD simulation only trains the classifier

# $t\bar{t}H$: the ($\kappa_b$, $\kappa_c$) plane

PRL 136 (2026) 011801




orange: observed, blue: expected; dashed 68%, solid 95% CL

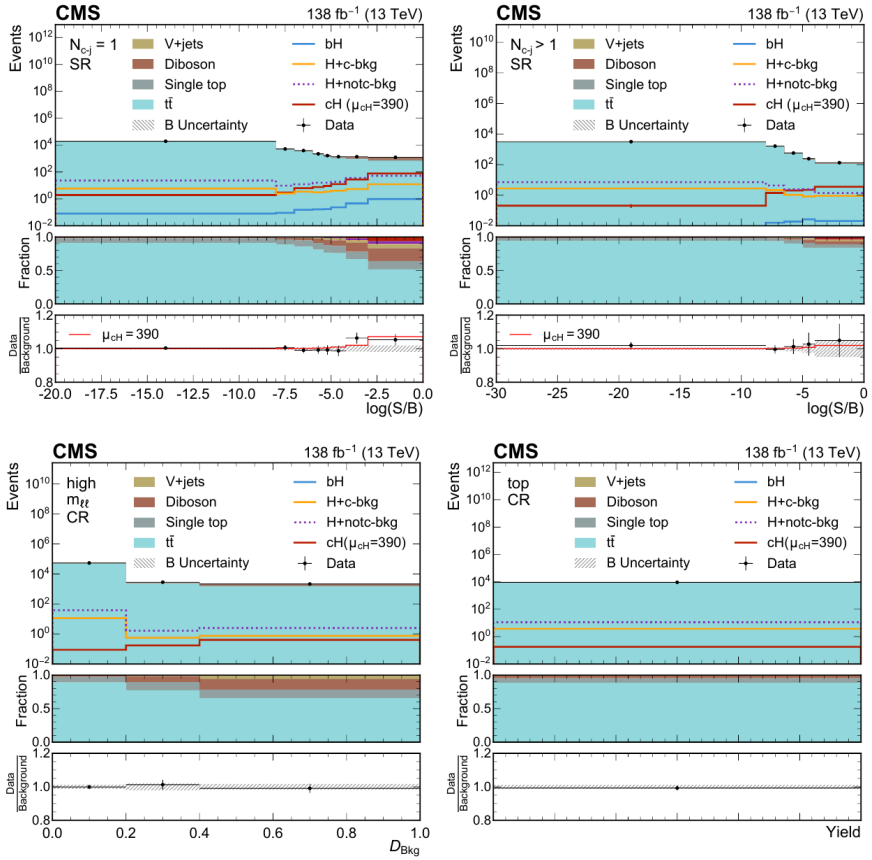
The simultaneous $t\bar{t}H(H \to b\bar{b})$ + $t\bar{t}H(H \to c\bar{c})$ fit constrains both Yukawas

Signal strengths: $t\bar{t}H(b\bar{b})$ $0.91\ ^{+0.26}_{-0.22}$ · $t\bar{t}H(c\bar{c})$ $-1.6 \pm 4.5$ · $t\bar{t}Z(b\bar{b})$ $1.47\ ^{+0.45}_{-0.41}$ · $t\bar{t}Z(c\bar{c})$ $1.02\ ^{+0.79}_{-0.84}$

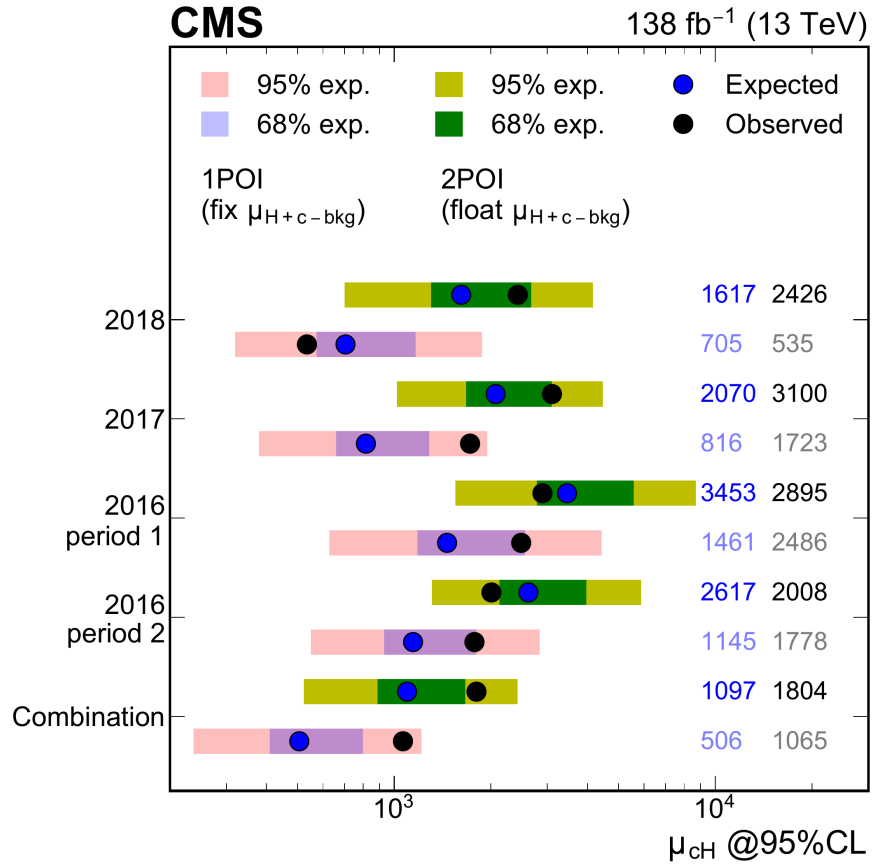
**$|\kappa_c| < 4.2$ obs. with $\kappa_b$ fitted**

# cH(H → WW): post-fit and per-period limits

JHEP 08 (2026) 051




1-POI post-fit BDT discriminants




$\mu_{cH}$ limits per data-taking period, 1 and 2 POI

➜ *Best fit (1 POI): $\mu_{cH} = 390\ ^{+380}_{-270}$, within 1.5σ of the SM; statistics-limited, then flavour-scheme and ggH heavy-flavour modelling*

# cH(H → WW): floating the H+c background

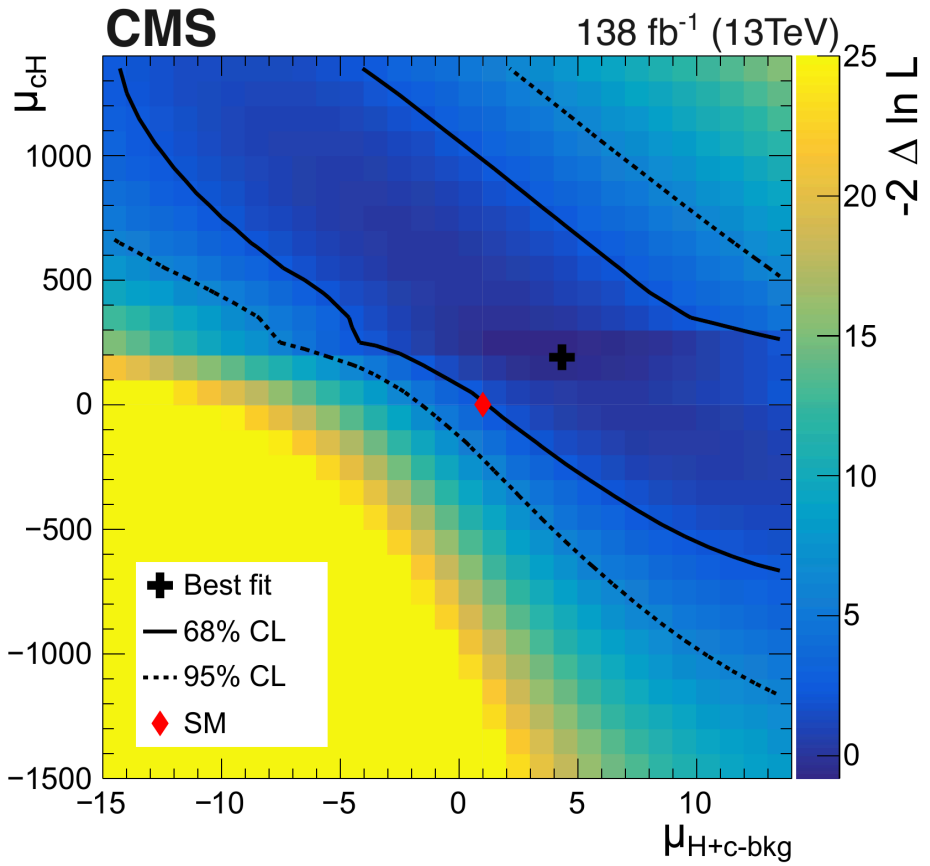
JHEP 08 (2026) 051

**1 POI** only $\mu_{cH}$ floats; the $\kappa_c$-independent H+c background (from ggH etc.) is fixed to its SM prediction

**2 POI** H+c background also floats; correlation with cH: −87%

2-POI limit: $\mu_{cH}$ < 1804 (1097)

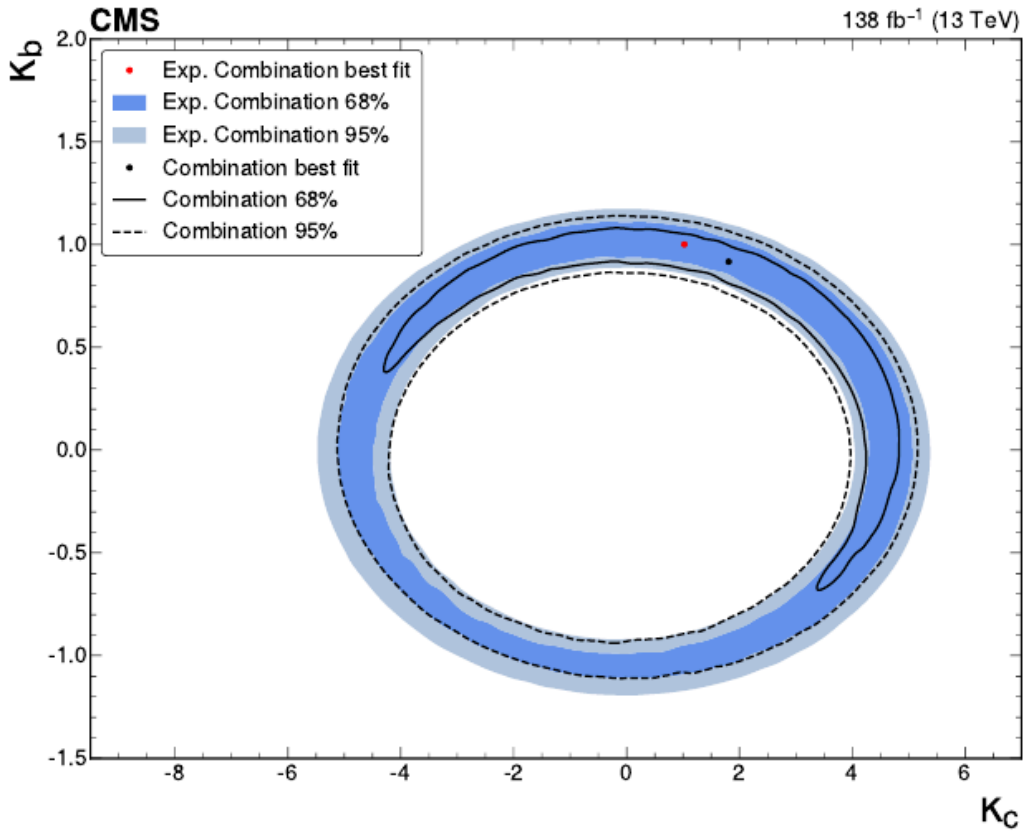
The combination with γγ uses the 1-POI fit



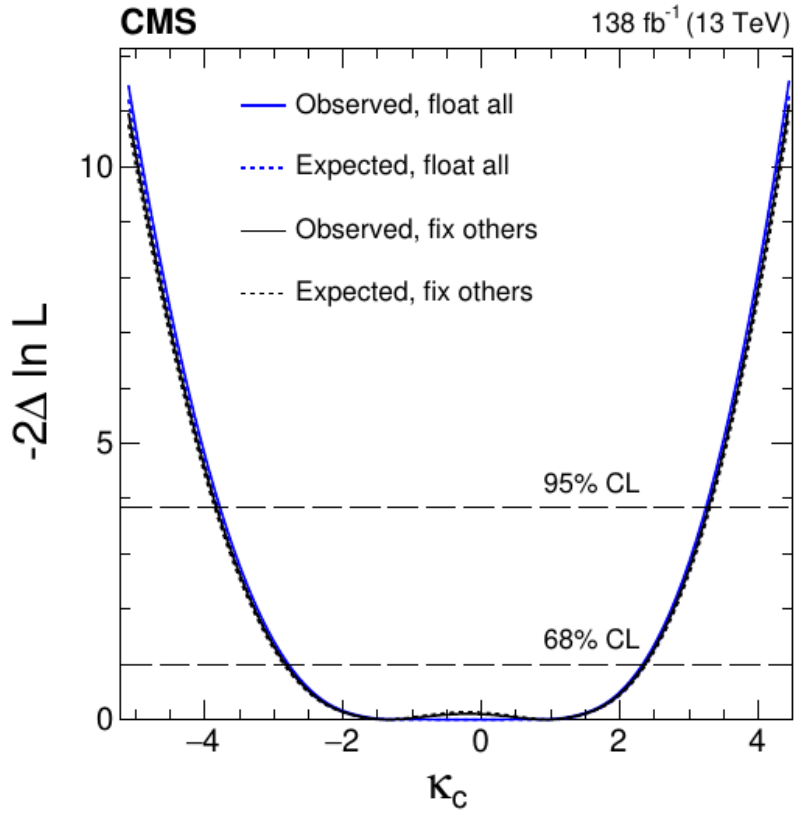

WW: $\mu_{cH}$ vs $\mu_{H+c\ bkg}$

***➜ Floating the H+c background weakens the WW limit from 1065 to 1804***

# Higgs $p_T$ spectrum and on-shell H → 4ℓ




JHEP 04 (2026) 093: ($\kappa_b$, $\kappa_c$) from the Run 2 $p_T^H$ combination (γγ, ZZ, resolved ττ), $\mathscr{B}$ coupling-dependent; 2D contours only




PRD 112 (2025) 112001 (γH + on-shell H → 4ℓ): $\kappa_c$ scan; solid obs., dashed exp.

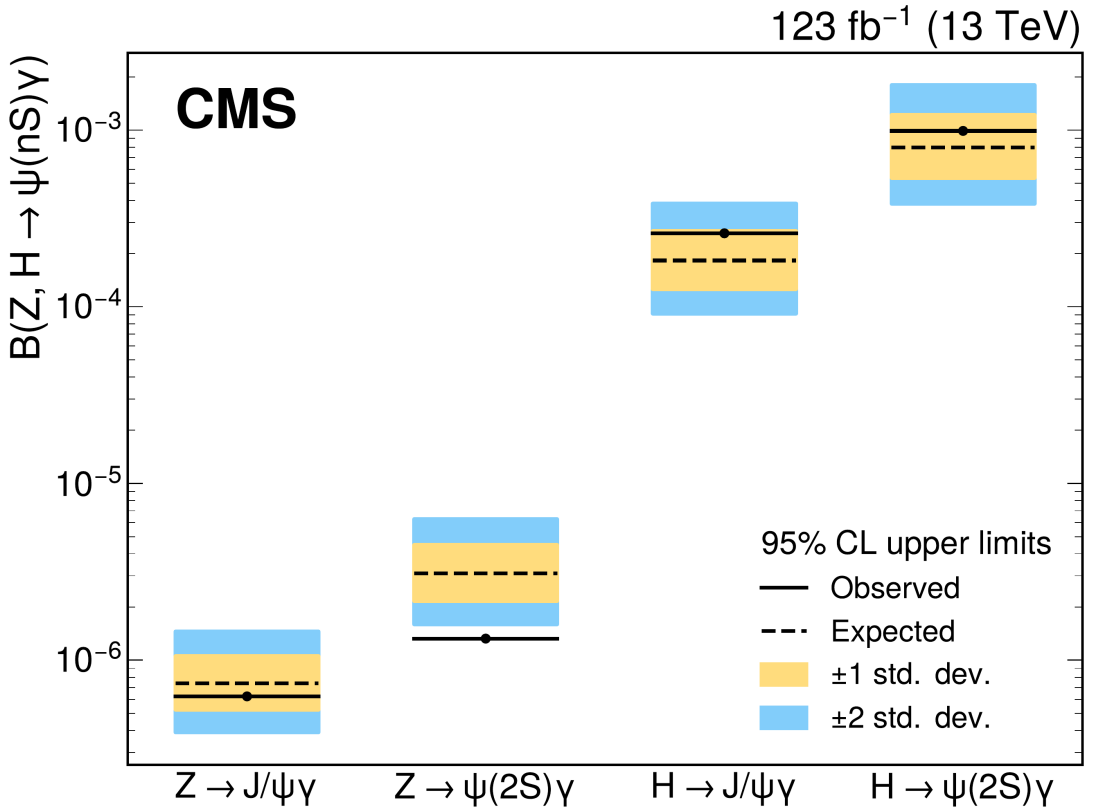
# Rare exclusive decays: H → J/ψ γ, ψ(2S) γ

PLB 865 (2025) 139462 · CMS-SMP-22-012

Direct ($\propto \kappa_c$) and indirect ($H \to \gamma\gamma^* \to \gamma\, J/\psi$) amplitudes interfere ⇒ **sign sensitivity**

123 fb$^{-1}$: $\mathcal{B}(H \to J/\psi\,\gamma) < 2.6 \times 10^{-4}$, $\mathcal{B}(H \to \psi(2S)\,\gamma) < 9.9 \times 10^{-4}$

**$\kappa_c/\kappa_\gamma \in (-157, 199)$**
**with $\kappa_\gamma = 1$: $\kappa_c \in (-166, 208)$**
expected (−128, 169) · 95% CL




95% CL upper limits on $\mathcal{B}(Z, H \to \psi(nS)\,\gamma)$, observed and expected

# H → μμ: a second-generation Yukawa

JHEP 01 (2021) 148 · CMS-HIG-19-006

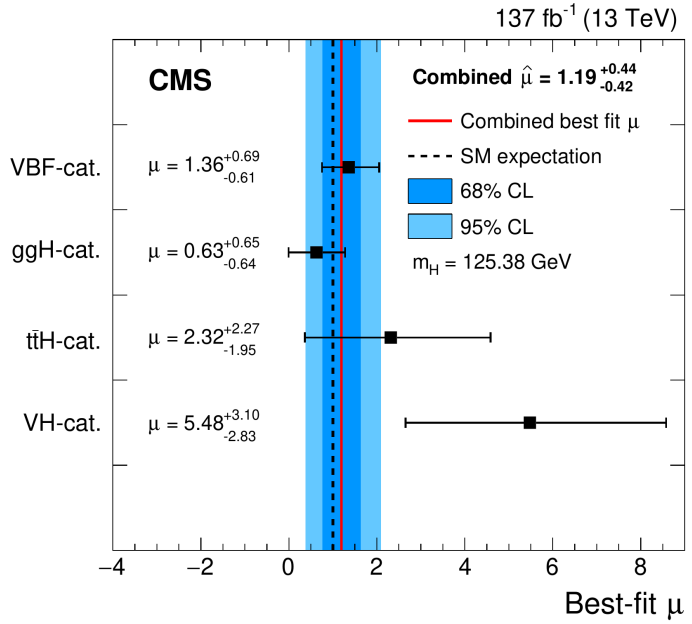
137 fb$^{-1}$, categories ggH, VBF, VH, $t\bar{t}H$

Excess of **3.0σ** (2.5σ exp.): first evidence for H decays to second-generation fermions

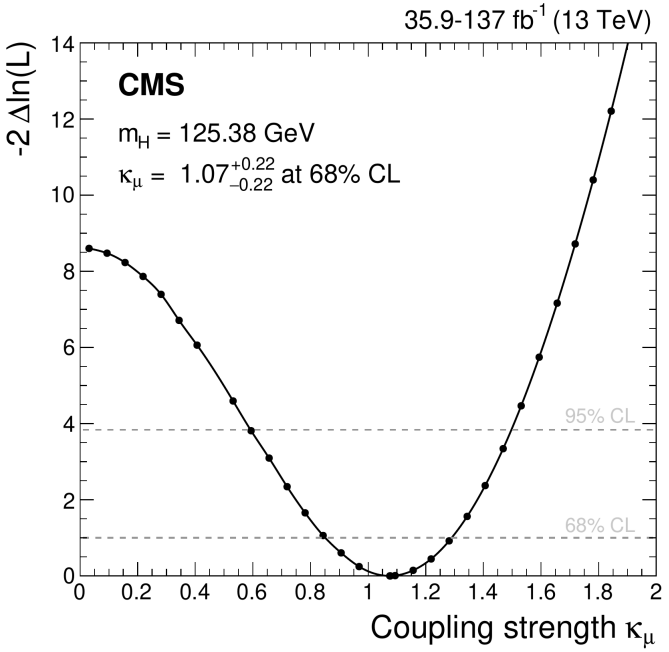
$\mu = 1.19\ ^{+0.40}_{-0.39}\ (\text{stat})\ ^{+0.15}_{-0.14}\ (\text{syst})$

$\kappa_\mu = 1.07$; 95% CL: $0.59 < \kappa_\mu < 1.50$ (6 free κ, SM total width)

Rare decays incl. μμ: L. D'Eramo's plenary (Wed)




μ̂ per production category and combined

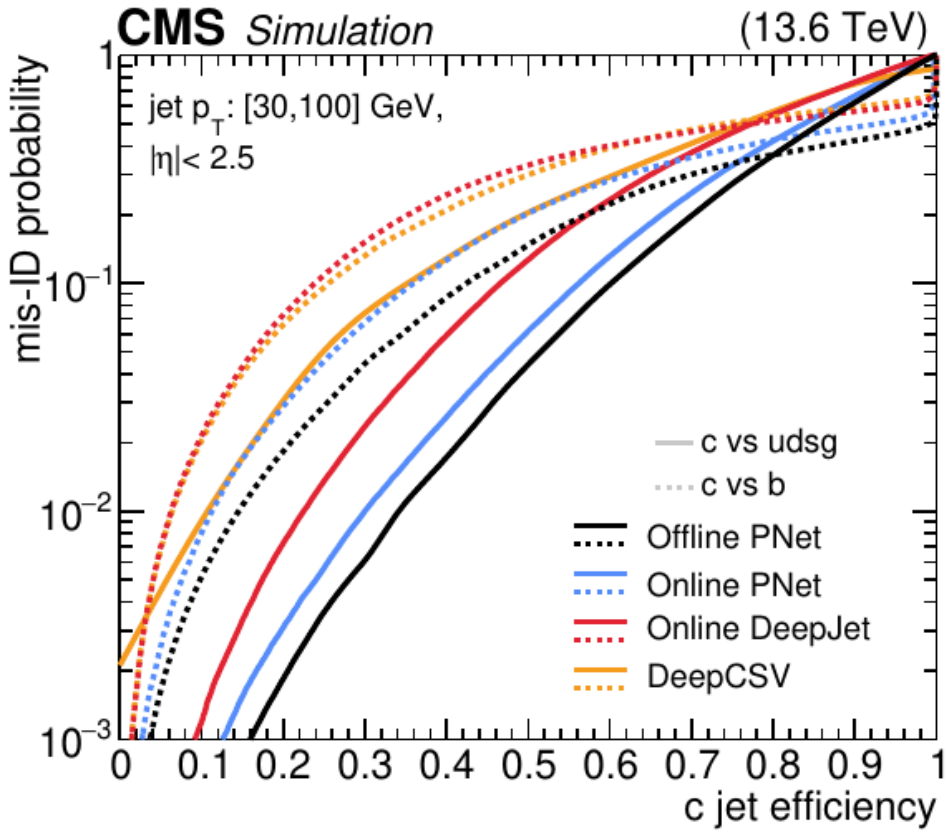



$-2\Delta\ln L$ vs $\kappa_\mu$

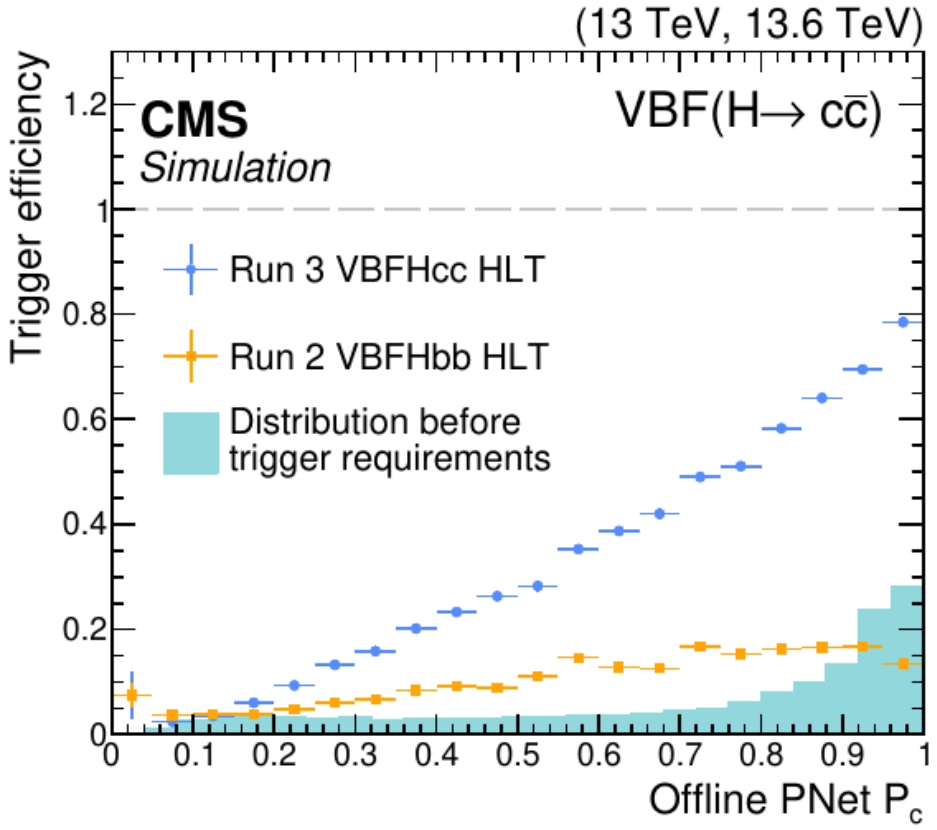
***➜ The muon Yukawa is SM-like within about 20% (68% CL); charm is the next test***

# Run 3 HLT flavour tagging

arXiv:2608.18331 · CMS-BTV-25-002




Simulation, jet $p_T$ 30–100 GeV: light (solid) and b (dashed) misidentification vs c-jet efficiency, online vs offline taggers




Simulation: VBF H → cc̄ trigger efficiency vs offline ParticleNet c score, Run 3 VBF H → cc̄ vs Run 2 VBF H → bb̄ trigger

# Likely questions

- **How large is the VH excess?** Best fit $\mu_{VH} = 7.7\ ^{+3.8}_{-3.5}$, about 1.9σ above μ = 1 (estimated from the quoted uncertainty; the paper calls it a small upward fluctuation). This is why VH alone excludes $|\kappa_c| < 1.1$ and why $t\bar{t}H$ alone (3.0) beats the combination (3.5) on observed
- **What limits the direct searches?** Statistics first. $t\bar{t}H(c\bar{c})$: statistical 74% of the total uncertainty, then $t\bar{t}$+c and $t\bar{t}$+light modelling (≈32%, ≈29%). $VH(c\bar{c})$: simulation sample size (≈37%), c-tagging efficiency (≈23%), theory (≈22%)
- **And ATLAS?** VBF + VH(H → $c\bar{c}$) (arXiv:2511.21911): $|\kappa_c| < 4.7$ (3.9 exp.), same assumptions as CMS (other couplings SM, no BSM decays); its VBF part includes 51.5 $fb^{-1}$ of Run 3 data
- **Why is cH an order of magnitude weaker than H → $c\bar{c}$?** cH $\sigma\mathscr{B}$ is tiny (≈0.2 fb in γγ) and ggH produces the same H+c final state independently of $\kappa_c$; in H → $c\bar{c}$, $\kappa_c$ changes the branching fraction of Higgs bosons from much larger production modes (VH, $t\bar{t}H$)
- **Why is cH(γγ) obs. (38) much better than exp. (73)?** The best fit is negative, $\hat{\mu}_{cH} = -230\ ^{+170}_{-251}$
- **Why no $\kappa_c$ from boosted H → $c\bar{c}$?** The paper quotes only μ; the result is ggH-dominated
- **How much of the Run 3 trigger gain is the tagger?** Part of it is rate budget: 10 Hz in 2018 vs 60, 115, 170 Hz in 2022, 2023, 2024
- **Are the H → 4ℓ intervals sign-sensitive?** Not really: the small asymmetry comes from charm-loop interference in ggH and H → gg; the constraint is essentially on $\kappa_c^2$

# References


- VH(H → $c\bar{c}$): CMS, PRL 131 (2023) 061801, arXiv:2205.05550 (HIG-21-008)
- $t\bar{t}$H(H → $c\bar{c}$) + VH combination: CMS, PRL 136 (2026) 011801, arXiv:2509.22535 (HIG-24-018)
- VH(H → $c\bar{c}$) with 2016 data (analysis strategy): CMS, JHEP 03 (2020) 131, arXiv:1912.01662 (HIG-18-031)
- Boosted H → $c\bar{c}$: CMS, PRL 131 (2023) 041801, arXiv:2211.14181 (HIG-21-012)
- cH(H → γγ): CMS, JHEP 11 (2025) 060, arXiv:2503.08797 (HIG-23-010)
- cH(H → WW) and combination: CMS, JHEP 08 (2026) 051, arXiv:2508.14988 (HIG-24-009)
- Differential combination Run 2: CMS, JHEP 04 (2026) 093, arXiv:2504.13081 (HIG-23-013)
- γH + H → 4ℓ light Yukawas: CMS, PRD 112 (2025) 112001, arXiv:2502.05665 (HIG-23-011)
- Off-shell + on-shell H → 4ℓ: CMS, arXiv:2607.23352, submitted to PRD (HIG-25-003)
- H → J/ψ γ: CMS, PLB 865 (2025) 139462, arXiv:2411.15000 (SMP-22-012)
- H → μμ: CMS, JHEP 01 (2021) 148, arXiv:2009.04363 (HIG-19-006)
- Jet flavour tagging in Run 3: CMS-DP-2025-081; DeepJet: E. Bols et al., JINST 15 (2020) P12012
- HLT flavour tagging: CMS, arXiv:2608.18331 (BTV-25-002)
- $t\bar{t}$+b/c cross sections: CMS-PAS-TOP-26-003
- ATLAS VBF + VH(H → $c\bar{c}$): ATLAS, arXiv:2511.21911
- HL-LHC: ATLAS and CMS, arXiv:2504.00672 (HIG-25-002)
- Higgs combination: CMS, Rep. Prog. Phys. 89 (2026) 087801, arXiv:2602.18611 (HIG-21-018)
- Spontaneous flavour violation: D. Egana-Ugrinovic, S. Homiller, P. Meade, PRL 123 (2019) 031802, arXiv:1811.00017; $\kappa_c$ in SFV: A. S. Giannakopoulou, P. Meade, M. Valli, "How charming can the Higgs be?", JHEP 02 (2025) 067, arXiv:2410.05236
- SM Higgs branching ratios: LHC Higgs Cross Section WG, CERN Yellow Report 4, CERN-2017-002-M, arXiv:1610.07922
- HL-LHC Yellow Report: M. Cepeda et al., CERN Yellow Rep. Monogr. 7 (2019) 221, arXiv:1902.00134